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Bearings in electric motors

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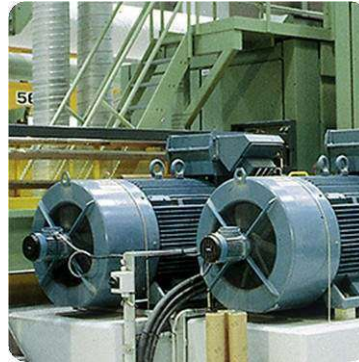
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Training objectives

Objectives:

- Learn fundamentals for entire course
- Learn best practices for enhancing electric motor mechanical reliability
- Understand common errors in electric motor repair and installation



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Electric motor chapter: objectives

- Purpose of motors
- Bearing basics
- Identify correct bearings
- Shaft and housing conformance
- Dismount bearings on cylindrical (straight) shafts
- Installation preparation
- Mount bearings on cylindrical (straight) shafts
- Taper Bore mounting and dismounting will be covered in the Industrial Fans section

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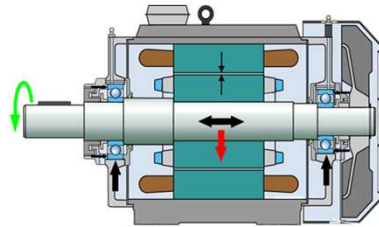
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Bearings in

Purpose:



-  Support the rotor
-  Locate the rotor
-  Keep air gap small and consistent
-  Transfer loads from the shaft to the motor frame
-  Allow both low and high speeds
-  Reduce friction and save power

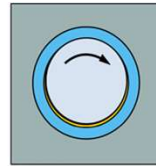
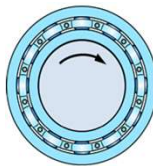
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Why use rolling element bearings?

Rolling bearings versus plain bearings



Friction torque, start	1	28–80
Friction torque, running	~0,5	~1,4
Lubrication, grease	Yes	No
Lubrication, oil	Yes	Yes
Vertical mounting	Yes	Special design
High speed	Yes	Yes
Low speed	Yes	No

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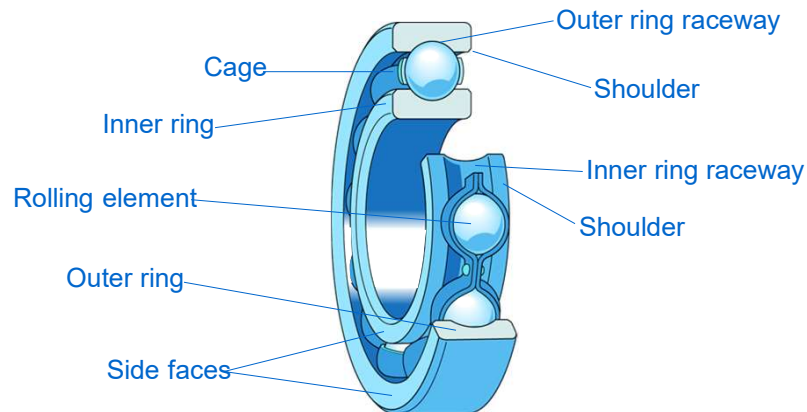
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Bearing basics: electric motor bearings

Bearing Components



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Bearing basics: electric motor bearings - cages

Examples of Cage Designs



Pressed sheet
steel cage



Polyamide 6,6
cage



Machined
brass cage

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Bearing basics: bearing designations

The basic designation identifies:

- Product type
- Standard design
- Size

The supplementary designation identifies:

- Alternative design (variants)
- Bearing components
- Types of special bearings

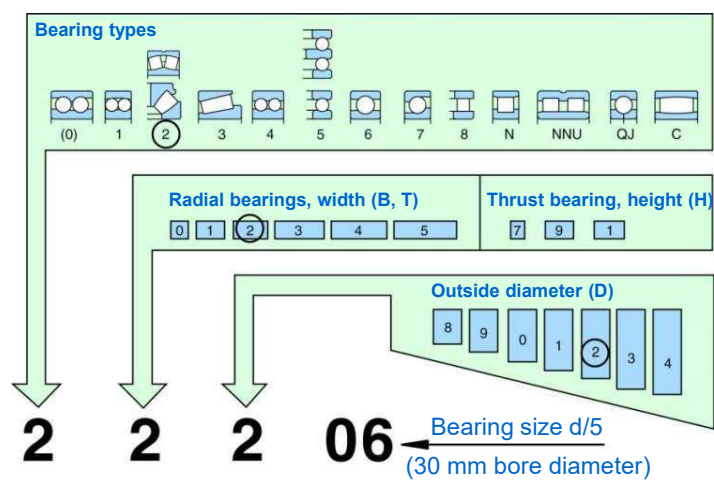
6210-2RS1JEM ?

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Bearing designations: how the system works



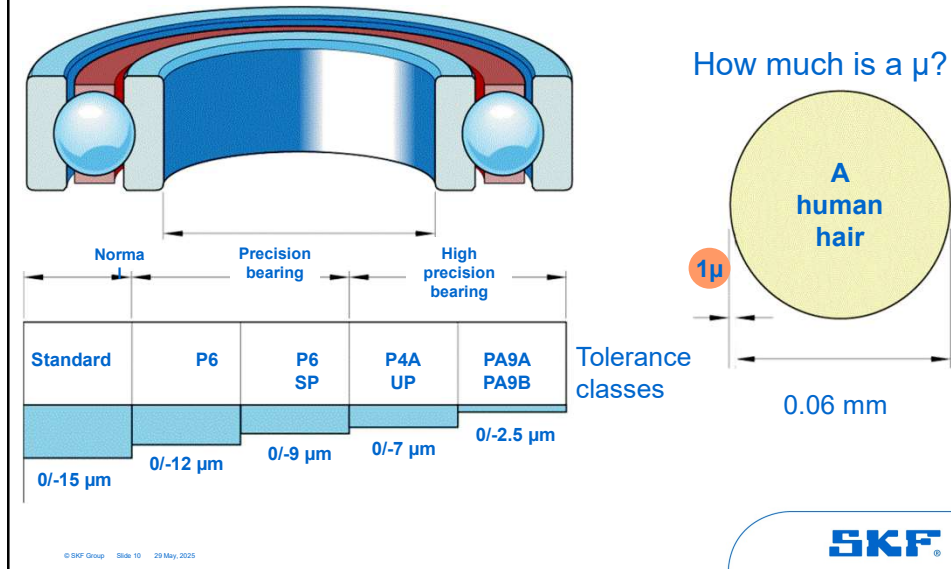
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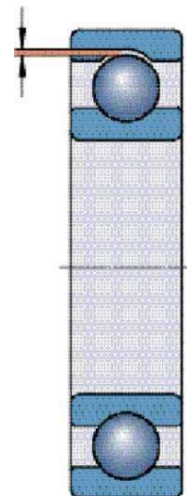
Bearing basics: bearing precision



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Bearing basics: internal clearance...

C1	Less than C2 clearance
C2	Less than normal clearance
(CN, C0)	Normal clearance (Not generally marked)
C3	Greater than normal clearance (Standard for electric motor ball bearings)
C4	Greater than C3 clearance
C5	Greater than C4 clearance



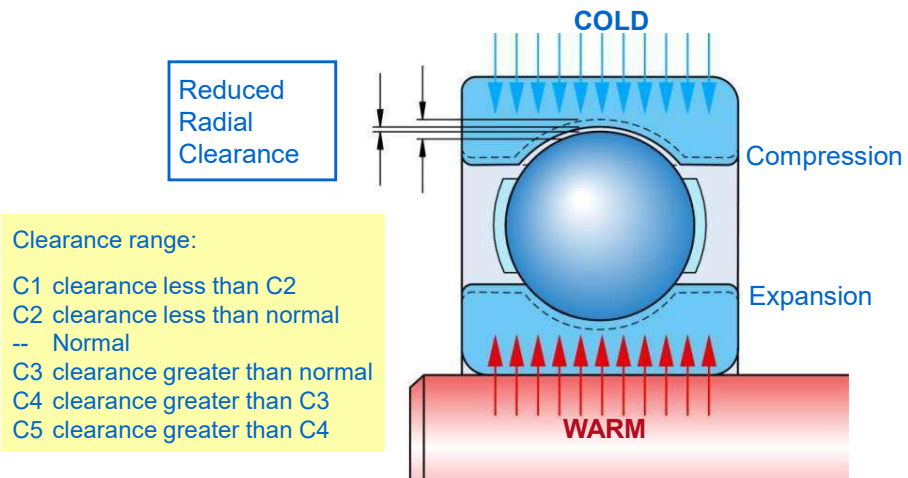
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Bearing basics: internal clearance...

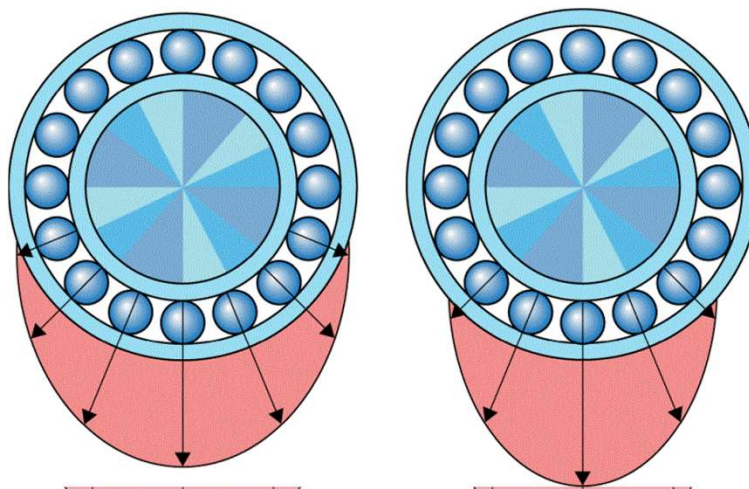


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Bearing basics: internal clearance



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Bearing arrangements

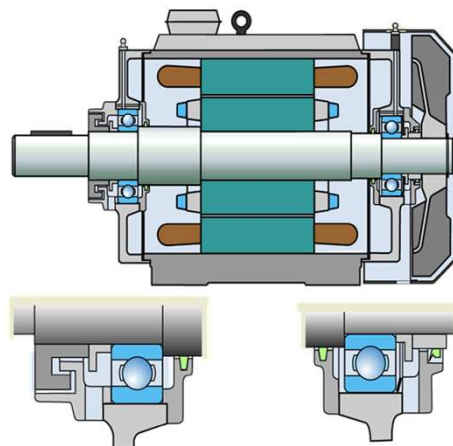
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Bearing arrangements: in

- One locating and one non-locating bearing or cross-locating arrangement
- Small motors mainly equipped with shielded/sealed deep groove ball bearings
- Medium to large motors are provided with re-lubrication means
- When frequent re-lubrication is needed, grease escape valves should be applied



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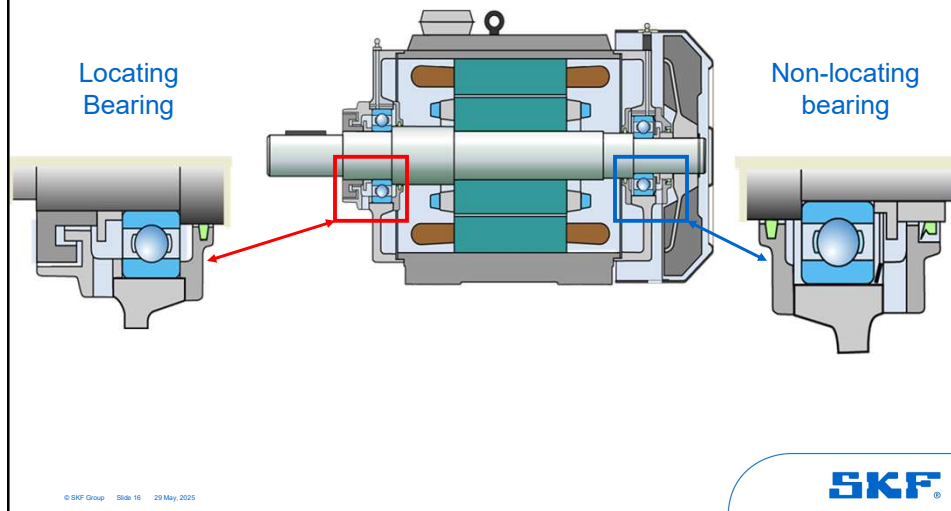
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Bearing arrangements: locating/non-locating

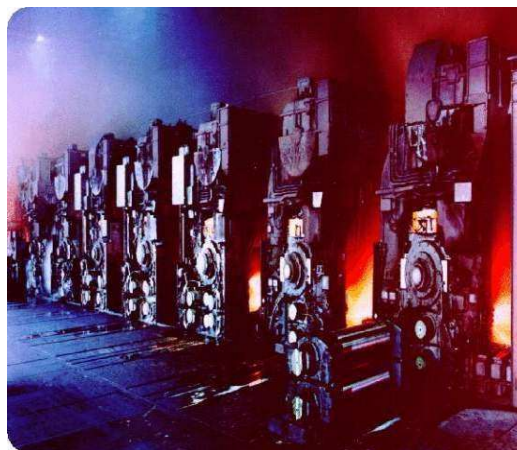
Locating or non-locating bearing?



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Thermal growth

- Steel shafts will grow with temperature
- What will happen if a shaft can't grow in any direction?



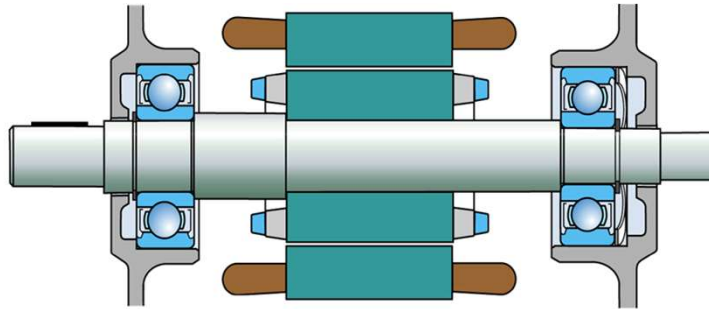
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Horizontal arrangement: small motors



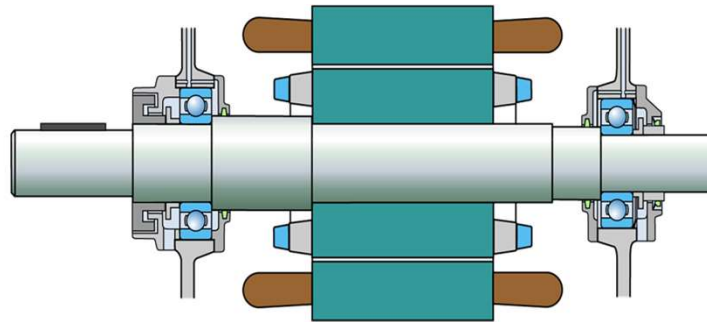
- Cross-locating arrangement with spring preload
- Axial guidance not key parameter
- Shielded bearings: Low maintenance

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Horizontal arrangement: medium size motors



- Two open deep groove ball bearings with grease escape valves
- High speed and low noise execution
- Non-locating bearing is spring preloaded
- Axial guidance not key parameter

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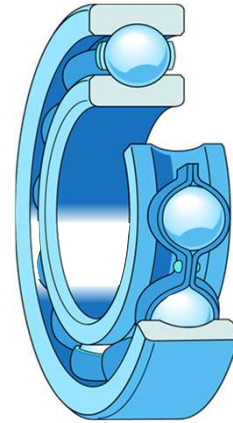
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Deep groove ball bearings: features

- Good radial and axial load carrying capacity, normally selected for small and medium sized motors
- Enable motor designs with spring force preloaded bearing arrangement, for high accuracy and silent running
- Sealed versions, greased for life = low maintenance
- Available in sealed version with the high performance, wide temperature grease WT



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Bearing basics: sample nomenclature...

Example: 6212-2Z/C3GJN

- 6** Single row deep groove ball bearing, Conrad style
- (0)2** “Light” dimension series
- 12** Multiply by 5 to get the bore in millimeters (works for bore designations 4-96)
- 2Z** Two shields
- C3** Greater than standard **radial** internal clearance
- GJN** Special SKF-approved electric motor grease

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Bearing basics: sample nomenclature

Example: 6320-2RS1JEM

- 6** Single row deep groove ball bearing, Conrad style
- (0)3** “Heavy” dimension series
- 20** Multiply by 5 to get the bore in millimeters
(works for bore designations 4-96)
- 2RS1** Two contacting seals
- JEM** **J** – steel cage, **EM** – Electric Motor Quality
 - **C3** - Greater than standard **radial** internal clearance
 - **GJN** - Special SKF-approved electric motor grease
 - **Q66** - Quiet running standard

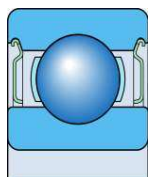
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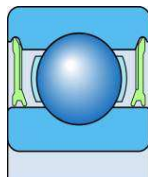
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Bearing basics: seals and shields

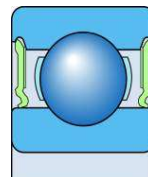
Deep groove ball bearings, sealed versions



Metallic
shields -ZZ



Low friction
seals -2RZ
-2RSL



Contact
seals -2RS1
-2RSH

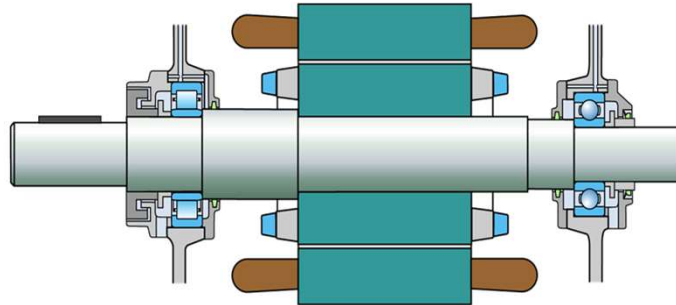
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Bearing arrangements: belt drive motors



- For motors with heavy loads on the drive side, belt drive, etc.
- Accommodates axial expansion within the bearing
- Equipped with grease valves
- Smaller motors, below frame size 132, usually equipped with two ball bearings

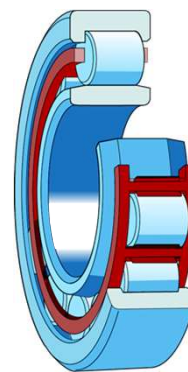
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Cylindrical roller bearings: features

- High radial load carrying capacity suitable for belt drive motors
- Excellent for the non-locating position, as it allows axial displacement inside the bearing
- Available in many different designs for both locating and non-locating positions



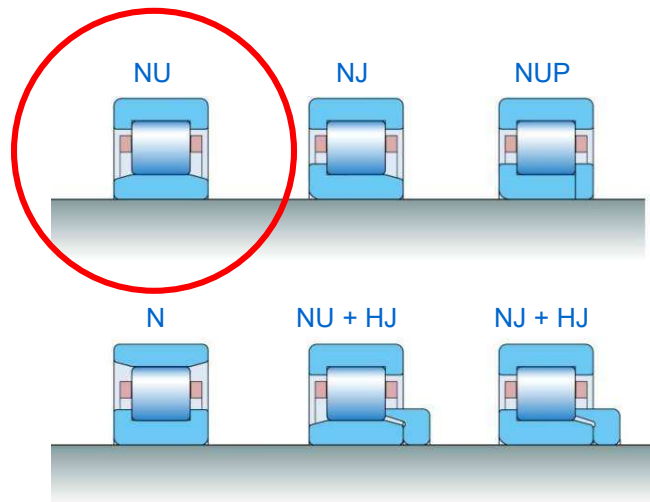
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Cylindrical roller bearings: versions



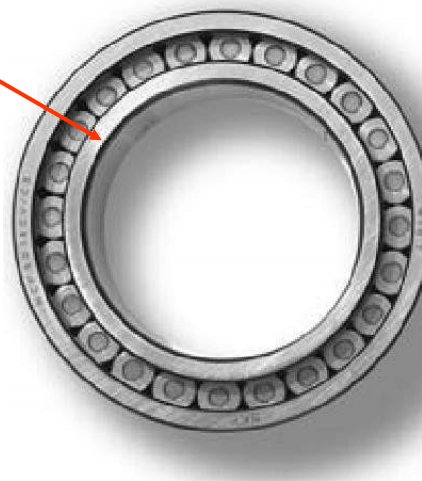
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Cylindrical roller bearings: designation

- Look at the inner ring for the correct part number
- Outer ring is a component number



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Converting from ball to roller bearings?

- Be careful, you are now changing the motor design! Liability?
- Call SKF/OEM engineering for support
- Always consider:
 - Different fits apply
 - Axial play
 - Proper location of the rotor

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CARB - Compact Aligning Roller Bearing

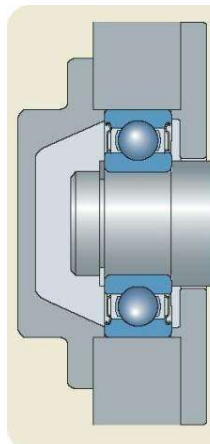
Great for:

- Belt drives
- Very high loads
- Misalignment

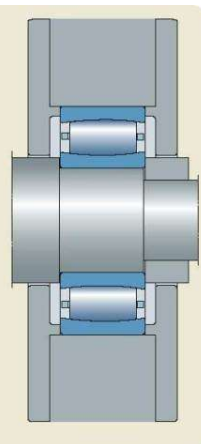
Allows for:

- Direct drives
- Shaft bending

Ball Bearing



CARB



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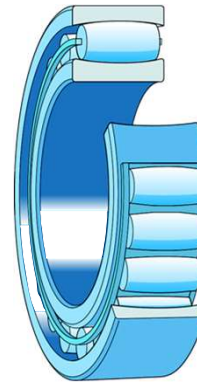
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CARB: features

- Very high radial load carrying capacity (suitable for belt drive motors)
- Allows for shaft misalignment
- Allows axial displacement inside the bearing
- Excellent for the non-locating position

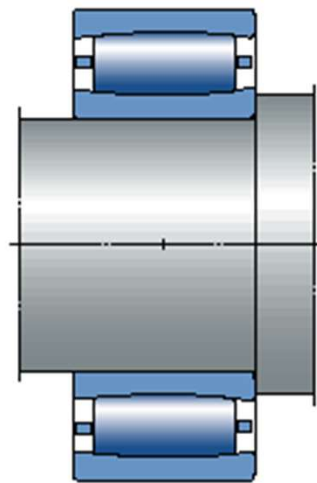


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Bearing basics: CARB...



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Bearing basics: CARB



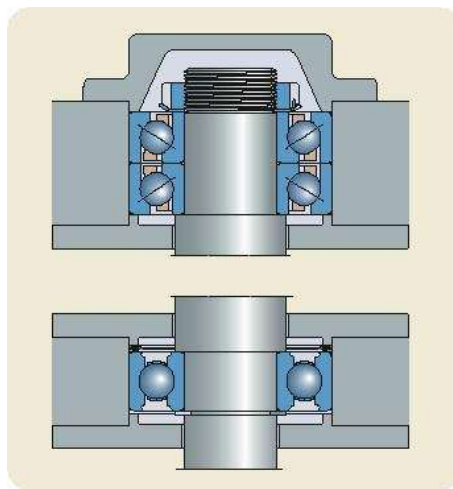
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Vertical electric motors: typical design

- Larger vertical electric motors have moderate axial loads in both directions
- Angular contact ball bearings are often used
- Back-to-back or face-to-face important design consideration; Do not alter!
- The shaft is also typically supported by a single row deep groove ball bearing



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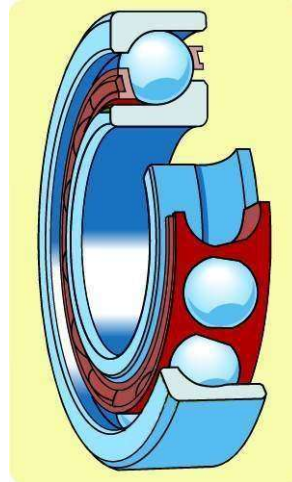
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Angular contact ball bearings: features

- Great for combined loads, axial (thrust), and radial loads
- High speed capacity
- Universally match-able for mounting in pairs
- Back-to-back, face-to-face, or tandem
- We will be returning to Angular Contact bearings in the “Pumps” section



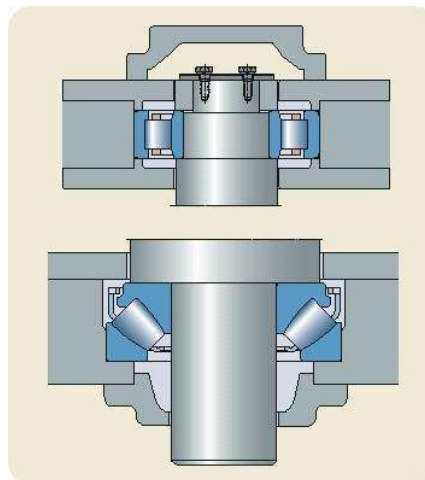
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Large vertical motors: typical design

- Large vertical motors often have heavy thrust loads
- Spherical roller thrust bearing will accept:
 - Heavy thrust loads
 - Dynamic misalignment (shaft deflection)
- Bearings are often spring preloaded for mounting and transportation
- Cylindrical roller bearing carries radial load



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Spherical roller thrust bearings: features

- Design allows very heavy axial (thrust) loads and radial loads
- Capable of relatively high speeds
- Self-aligning capability allows for full load under misalignment
- Separable design
- Often preloaded or used in pairs
- Internal pumping action for oil bath lubrication



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Variable speed motors

- The use of frequency converters is increasing
 - Energy saving in the range 15-40%
 - Flexibility, low on maintenance, robust
- Modern converters cause stray currents that damage the bearings
- Converters are getting more and more efficient, but worse for the bearings



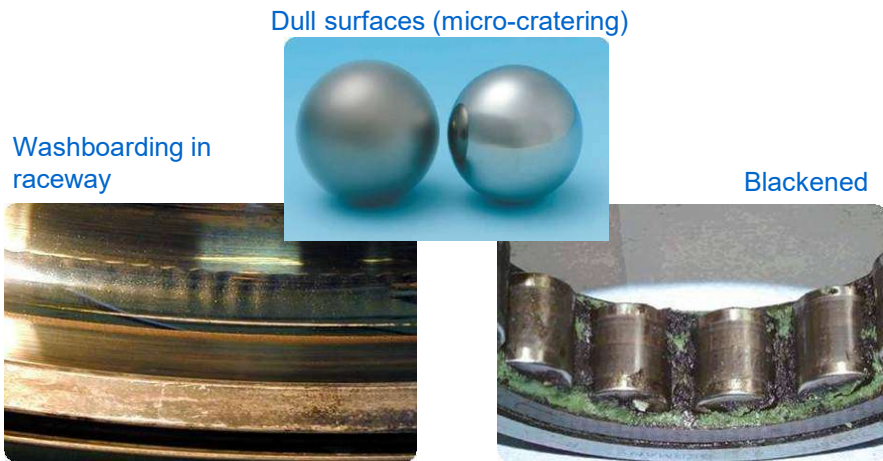
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Electric arcing/fluting



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Electric arcing/fluting/electrical erosion



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Insulated bearings: features

INSOCOAT Deep Groove Ball Bearing



Outer ring coated with an electrically insulating coating

Hybrid Deep Groove Ball Bearing



Rolling elements of ceramic material

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INSOCOAT: insulated bearing

- ISO boundary dimensions – drop in replacement
- Hot or cold mounting – tested for up to P6 fits
- Plasma spraying gives the highest coating quality
- Not sensitive to humidity, chemicals, temperature (most sleeves are)

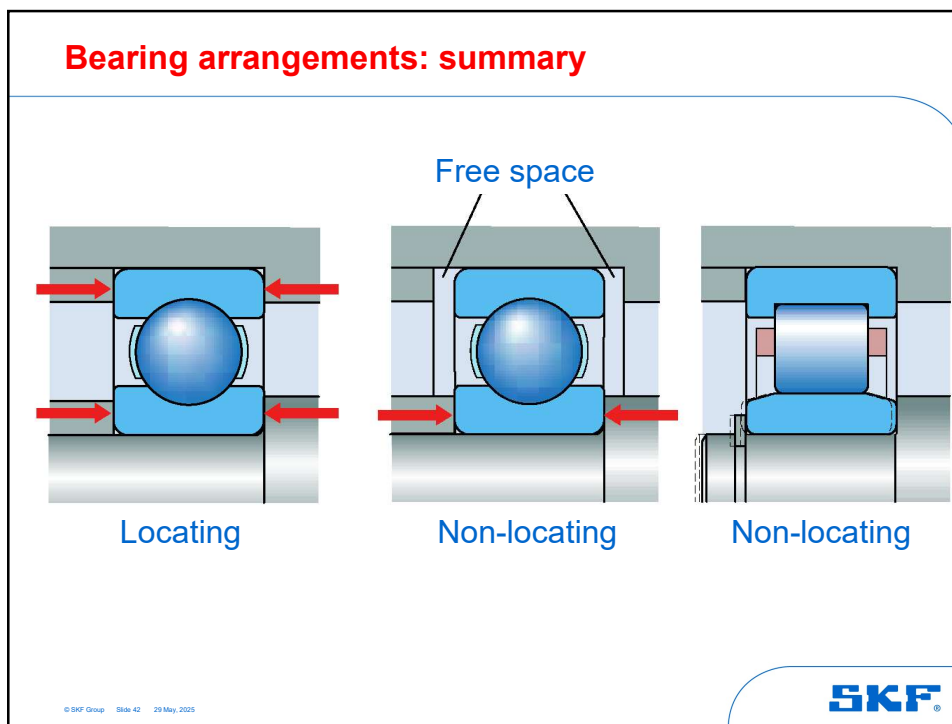


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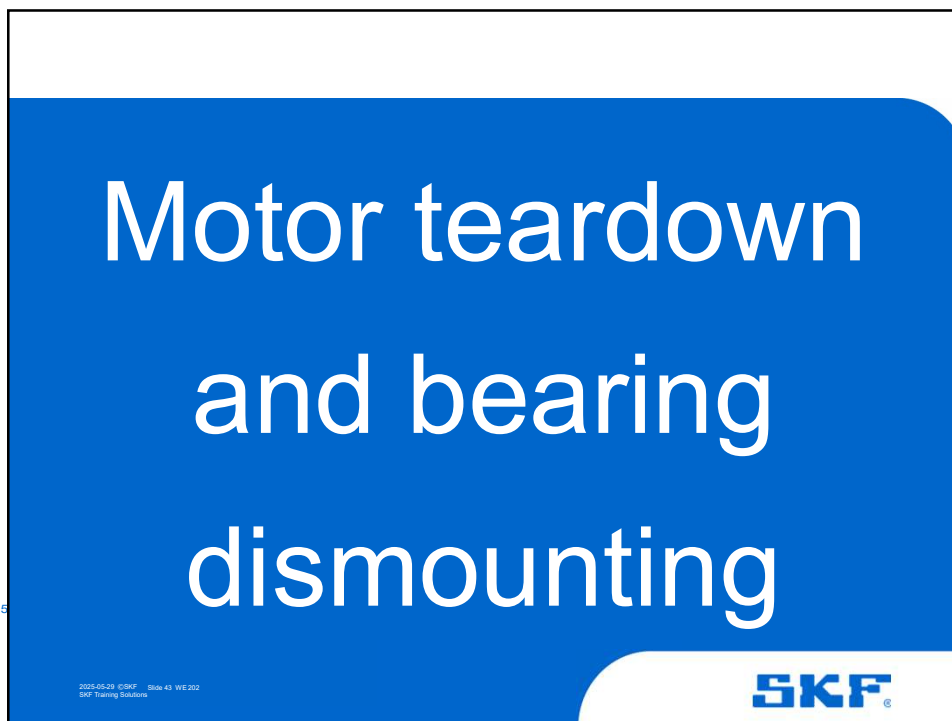
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Motor teardown and inspection

Assess the mechanical integrity of the motor

- Record full nameplate data, date, serial number, frame size, mounting configuration, etc.
- Mark and document, preferably with digital camera, all included accessories, such as terminal box, coupling, sheaves, etc.
- Measure and document bearing and seal fits
- Check shaft for mechanical defects, cracks, run-out, etc.
- Identify and document cause of failure (if any)
- Document with an “incoming inspection” report

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Identifying bearings for replacement...

If you know equipment is going out of service, it's possible to identify replacements before teardown

- Check equipment nameplate (ABMA)
- Check application print or contact manufacturer
- Check the EASA technical manual

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Identifying bearings for replacement

Important to know the exact bearing designation

- Look at side faces for part number
- Or consult the manufacturer's print
- Or call manufacturer for information
- Your distributor can usually help with cross-references
- SKF has excellent interchange resources

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Important to know the exact bearing designation

Warning!

If you don't know the exact designation and only the bearing dimension, you will not get any special features, such as:

- Clearance (!)
- Precision
- Bearing type (Deep Groove BB or Angular Contact BB?)
- Lubricant
- Other variations

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Hands-on activity #1

Hands-on activity:

- Identify the bearing that goes with each shaft
- Follow the Hands-on Activity Sheet #1

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Shaft and housing fits

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Shaft and housing fits

- Bearings are made with extreme precision for good reasons
- If a shaft is out of roundness or size, what will change shape?
- The shaft or the bearing ring?
- Improper fits can have a detrimental effect on motor service life



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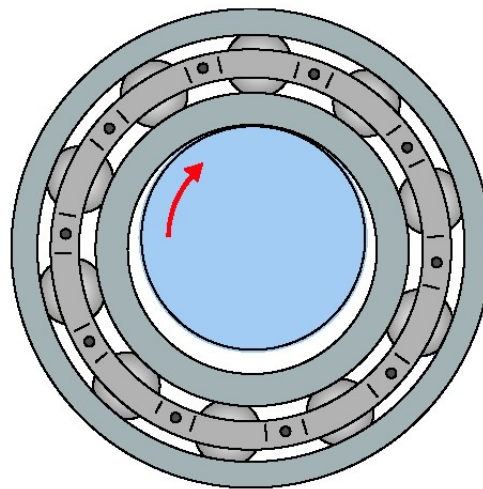
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Shaft and housing fits: importance of correct fit...

Excessively loose fit:

- Allows relative movement between mating components
 - Heat
 - Wear
 - Fretting
 - Smearing
 - Scoring
 - Ring fracture



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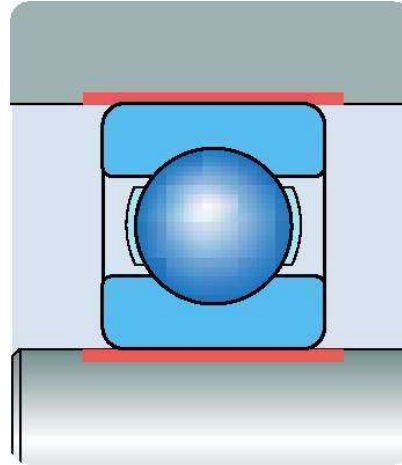
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Shaft and housing fits: importance of correct fit

Excessively tight fit:

- Creates reduction of internal clearance
 - Increased operating temperature
 - Ring fracture
 - Increased mounting forces



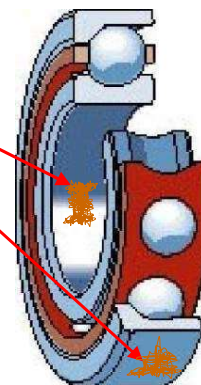
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Shaft and housing fits: checking fit integrity

- Remove the bearing
- Note the condition of O.D. and bore
- Examine for fretting corrosion:
 - Reddish-brown or black color
 - Nearly always present on the bore of housing inserts
 - Improper fit and rotating load produces tiny metal particles
 - Particles combine with air to create a form of rust
 - Technically known as fretting corrosion



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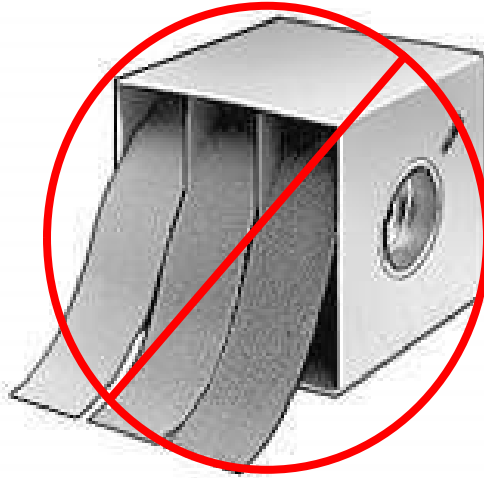
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Shaft and housing fits...

Clean up gently

- No emery
- No sandpaper
- No steel wool / wire wool



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Shaft and housing fits...

Don't take metal away

- Try a rag and solvent first
- Advance to a mild abrasive



Crocus Cloth

If fretting is excessive or chronic:

- Look for the cause of the problem
- Start with the SKF Troubleshooting Guide

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Shaft and housing fits

For nicks and burrs:

- Use a honing stone
- Never use a file

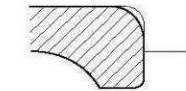
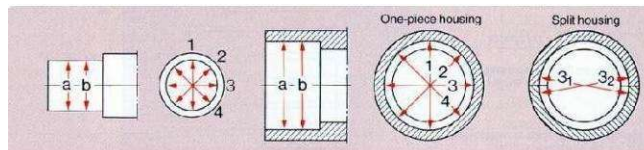


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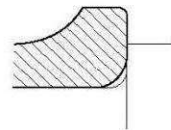
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Shaft and housing fits: mounting preparations...



Chamfer radius must be less than bearing corner radius



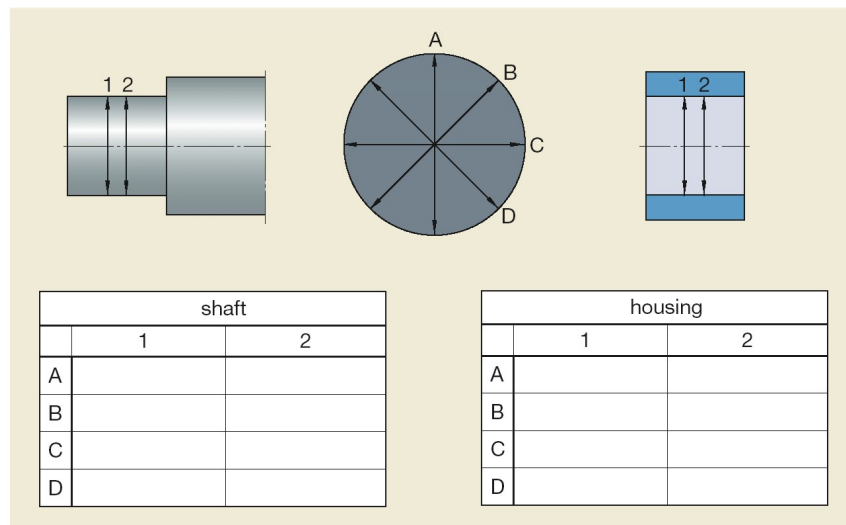
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Shaft and housing fits: mounting preparations



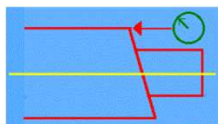
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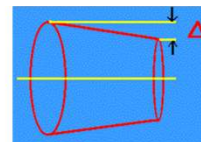
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Shaft and housing fits: form tolerances

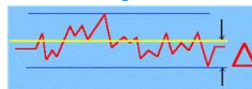
Shoulder squareness
(perpendicularity)



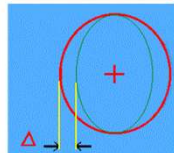
Taper (cylindricity)



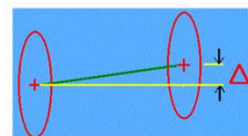
Roughness



Out-of-round (circularity)



Concentricity (coaxiality)



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Bearing seat run-out and form tolerance

The form of the seat is even more important than the size

- Should be half the size tolerance. Make sure to carefully check form tolerances, taper, etc.
- Check and document bearing seat run-out. High run-out will affect bearing and motor life negatively

Maximum Bearing Seat Run-out

Diameter da/Da (mm)		Form tolerance inches (um) IT5 ¹
Over	Including	
10	50	0.0003" (8um)
50	120	0.0005" (13um)
120	250	0.0007" (18um)
250	-	0.0009" (23um)

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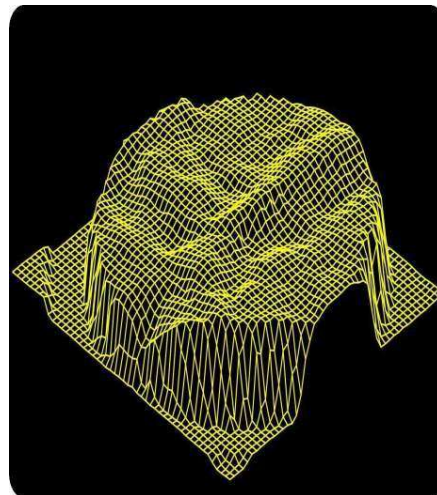
60

Surface roughness: bearing seats

Surface roughness has limited influence on bearing performance compared to:

- Form
- Run-out

$R_a = 32$ micro-inches is sufficient (fine machining, but not grinding)



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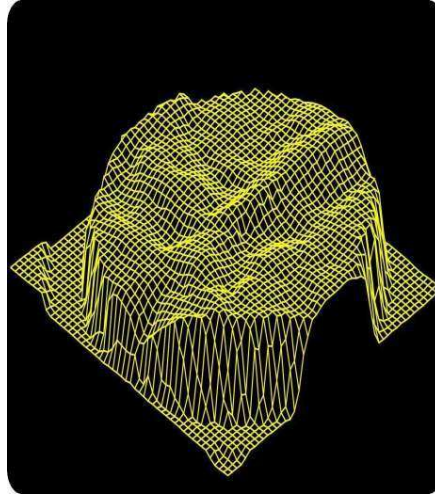
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Surface roughness: seal seats

Surface roughness has great impact on seal performance

- $R_a = 10\text{-}20$ micro-inches
- Plunge ground, with a machine lead of 0 ± 3 minutes
- Check “Dynamic run-out” (DRO) and tolerance requirements
- “Speedi-sleeves” often an attractive alternative

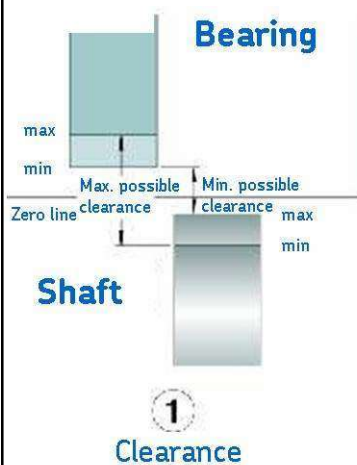


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Shaft and housing fits: examples of fit types...



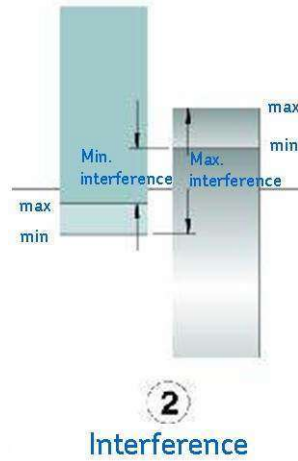
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Shaft and housing fits: examples of fit types...

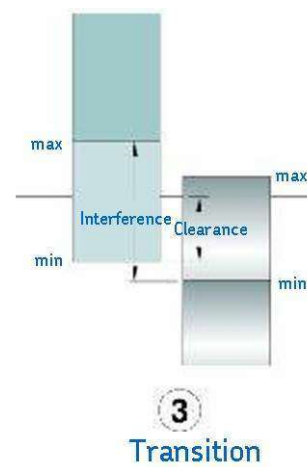


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Shaft and housing fits: examples of fit types...

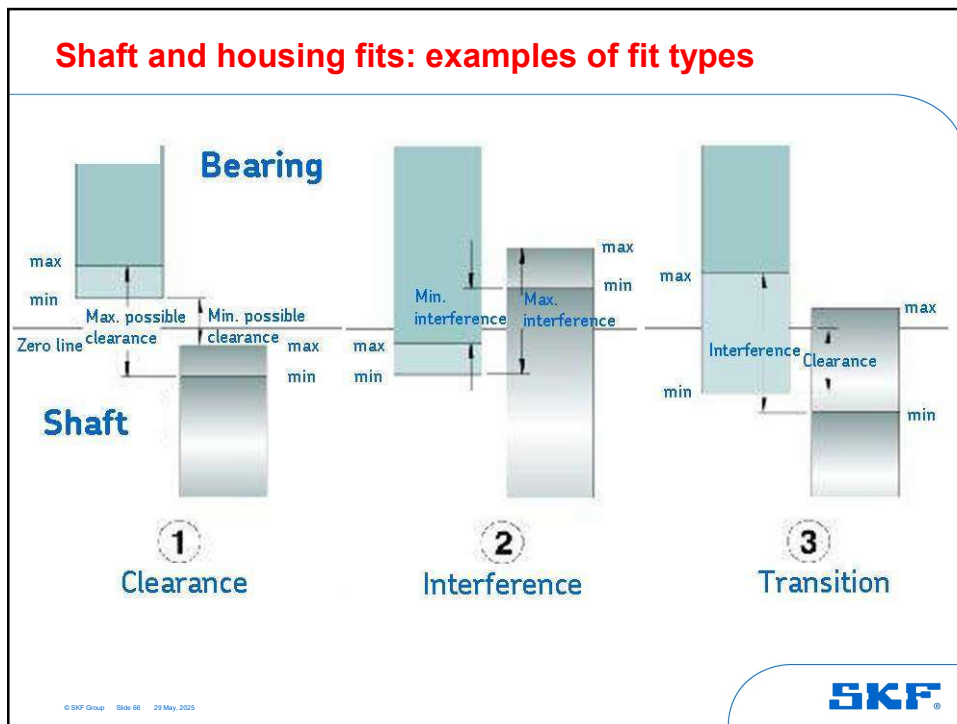


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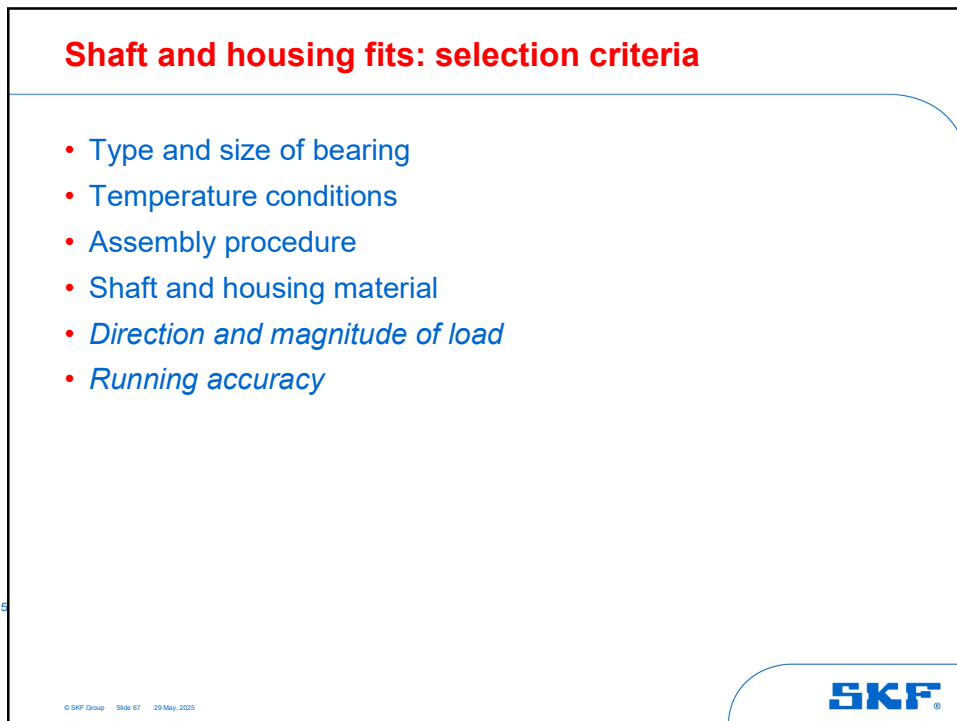
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Shaft and housing fits...

SKF Bearing Installation and Maintenance Guide topics:

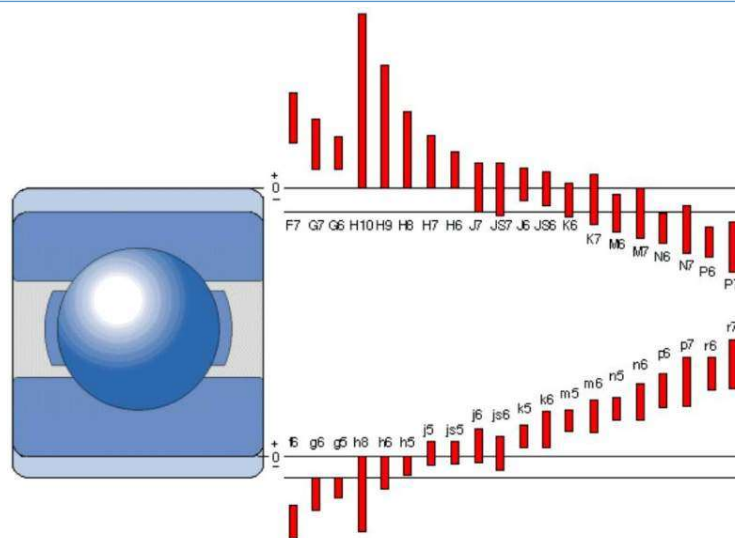
- Bearing types and terminology
- Shaft and housing fits
- Lubrication
- Troubleshooting and failure analysis
- Training resources

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Shaft and housing fits



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Shaft and housing fits: selection of tolerance...

Bearing number: 6212 2RS1JEM
Application: electric motor
Rotation: inner ring
Bearing type: ball
Bore size: 60 mm
Tolerance symbol: ???

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Shaft and housing fits: selection of tolerance

- Right fit for this specific motor type?
 - Consult the motor OEM to get the correct tolerance symbol (fit)
 - If not possible, ask your customer for documentation of fits
- SKF has recommended fits if you cannot get the information from the OEM or your customer

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Selection of tolerance...

Selection of Solid Steel Shaft Tolerance
Classification for Metric Radial Ball and Roller Bearings of Tolerance
Classes ABEC-1, RBEC-1
(Except Inch Dimensioned Tapered Roller Bearings)

Conditions	Shaft diameter, mm Ball bearings ¹	Cylindrical roller bearings, metric taper	Spherical roller bearings	Tolerance Symbol
Rotating inner ring load or direction of loading indeterminate				
Light loads	18≤100 100≤140	≤40 40≤100	— —	j6 k6
Normal loads	≤18 18≤100 100≤140 140≤200 200≤280 — — —	— ≤40 40≤100 100≤140 140≤200 200≤400 — — —	— ≤40 40≤65 65≤100 100≤140 140≤280 280≤500 >500	i6 k5 (k6) ² m6 (m6) ² m6 n6 p6 r6 r7
Heavy loads	— — —	50≤140 140≤200 ≤200	50≤100 100≤140 >140	n6 ³ p6 ³ r6 ³
High demands on running accuracy with light loads	≤18 18≤100 100≤200 —	— ≤40 40≤140 140≤200	— — — —	h5 ⁴ j5 ⁴ k5 ⁴ m5 ⁴

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Selection of tolerance

Shaft Bearing-Seal Diameters (Values in inches) (cont.)

Bearing Diameters (mm)	Bore (inches)		j6		Fit in 0.0000"	k5		Fit in 0.0000"	k6		Fit in 0.0000"	m5		Fit in 0.0000"	m6		Fit in 0.0000"
	Max.	Min.	Max.	Min.		Max.	Min.		Max.	Min.		Max.	Min.		Max.	Min.	
4	0.1575	0.1572	0.1577	0.1574		0.1577	0.1575		0.1579	0.1575		0.1579	0.1577		0.1580	0.1577	
5	0.1969	0.1966	0.1971	0.1968	1 L	0.1971	0.1969	7 T	0.1973	0.1969	0 T	0.1973	0.1971	2 T	0.1974	0.0001	2 T
6	0.2362	0.2359	0.2364	0.2361	5 T	0.2364	0.2362	5 T	0.2366	0.2362	7 T	0.2366	0.2364	7 T	0.2367	0.2364	8 T
7	0.2756	0.2753	0.2759	0.2755		0.2759	0.2756		0.2760	0.2756		0.2761	0.2758		0.2762	0.2758	
8	0.3150	0.3147	0.3153	0.3149	1 L	0.3153	0.3150	0 T	0.3154	0.3150	0 T	0.3155	0.3152	2 T	0.3156	0.3152	2 T
9	0.3543	0.3540	0.3546	0.3542	6 T	0.3546	0.3543	6 T	0.3547	0.3543	7 T	0.3548	0.3545	8 T	0.3549	0.3545	9 T
10	0.3937	0.3934	0.3940	0.3936		0.3940	0.3937								0.3939		
12	0.4724	0.4721	0.4727	0.4723		0.4728	0.4724								0.4727		
15	0.5906	0.5903	0.5909	0.5905	1 L	0.5910	0.5906	0 T							0.5909		3 T
17	0.6693	0.6690	0.6696	0.6692	6 T	0.6697	0.6693	7 T	0.6699	0.6695	8 T	0.6699	0.6696	9 T	0.6700	0.6696	10 T
20	0.7874	0.7870	0.7878	0.7872		0.7878	0.7875		0.7880	0.7875		0.7881	0.7877		0.7882	0.7877	
25	0.9843	0.9839	0.9847	0.9841	2 L	0.9847	0.9844	1 T	0.9849	0.9844	1 T	0.9850	0.9846	3 T	0.9851	0.9846	3 T
30	1.1811	1.1807	1.1815	1.1809	8 T	1.1815	1.1812	8 T	1.1817	1.1812	10 T	1.1818	1.1814	11 T	1.1819	1.1814	12 T
35	1.3780	1.3775	1.3784	1.3778		1.3785	1.3781		1.3787	1.3781		1.3788	1.3784		1.3790	1.3784	
40	1.5748	1.5743	1.5752	1.5746	2 L	1.5753	1.5749	1 T	1.5755	1.5749	1 T	1.5756	1.5752	4 T	1.5758	1.5752	4 T
45	1.7717	1.7712	1.7721	1.7715	9 T	1.7722	1.7718	10 T	1.7724	1.7718	12 T	1.7725	1.7721	13 T	1.7727	1.7721	15 T
50	1.9685	1.9680	1.9689	1.9683		1.9690	1.9686		1.9692	1.9686		1.9693	1.9689		1.9695	1.9689	
55	2.1654	2.1648	2.1658	2.1651		2.1659	2.1655		2.1662	2.1655		2.1663	2.1658		2.1666	2.1658	
60	2.3622	2.3615	2.3626	2.3619		2.3628	2.3623		2.3630	2.3623		2.3631	2.3626		2.3634	2.3626	
65	2.5591	2.5585	2.5595	2.5588	3 L	2.5597	2.5592	1 T	2.5599	2.5592	1 T	2.5600	2.5595	4 T	2.5603	2.5595	4 T
70	2.7560	2.7553	2.7563	2.7556	11 T	2.7565	2.7560	12 T	2.7567	2.7560	14 T	2.7568	2.7563	15 T	2.7571	2.7563	18 T
75	2.9528	2.9522	2.9532	2.9525		2.9534	2.9529		2.9536	2.9529		2.9537	2.9532		2.9540	2.9532	
80	3.1496	3.1490	3.1500	3.1493		3.1501	3.1496		3.1504	3.1497		3.1505	3.1500		3.1508	3.1500	
85	3.3465	3.3458	3.3469	3.3462		3.3471	3.3465		3.3473	3.3465		3.3475	3.3470		3.3478	3.3470	
90	3.5433	3.5425	3.5437	3.5429		3.5440	3.5433		3.5443	3.5434		3.5444	3.5438		3.5447	3.5438	
95	3.7402	3.7394	3.7406	3.7397		3.7409	3.7402		3.7412	3.7403		3.7413	3.7407		3.7416	3.7407	
100	3.9370	3.9361	3.9374	3.9365		3.9377	3.9369		3.9380	3.9371		3.9381	3.9375		3.9384	3.9375	
105	4.1339	4.1331	4.1344	4.1335	4 L	4.1346	4.1340	15 T	4.1349	4.1340	18 T	4.1350	4.1344	19 T	4.1353	4.1344	22 T
110	4.3307	4.3299	4.3312	4.3303		4.3314	4.3308		4.3317	4.3308		4.3318	4.3312		4.3321	4.3312	
115	4.5276	4.5268	4.5281	4.5272		4.5283	4.5277		4.5286	4.5277		4.5287	4.5281		4.5290	4.5281	
120	4.7244	4.7236	4.7249	4.7240		4.7251	4.7245		4.7254	4.7245		4.7255	4.7249		4.7258	4.7249	

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Shaft and housing fits: selection of tolerance

2.3628 – 2.3623

2.3628

2.3623

.0005

Out of roundness and taper should be half the tolerance

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Selection of tolerance

Shaft Bearing-Seat Diameters (Values in inches) (cont.)

Bearing Diameters (mm)	Bore (inches)		j5			k5			k6			m5			m6		
	Max.	Min.	Shaft Dia. Max.	Min.	Fit in 0.0000"	Shaft Dia. Max.	Min.	Fit in 0.0000"	Shaft Dia. Max.	Min.	Fit in 0.0000"	Shaft Dia. Max.	Min.	Fit in 0.0000"	Shaft Dia. Max.	Min.	Fit in 0.0000"
4	0.1575	0.1572	0.1577	0.1574		0.1577	0.1575		0.1579	0.1575		0.1579	0.1577		0.1580	0.1577	
5	0.1969	0.1966	0.1971	0.1968	1 L	0.1971	0.1969		0.1973	0.1969	0 T	0.1973	0.1971	2 T	0.1974	0.0001	2 T
6	0.2362	0.2359	0.2364	0.2361	5 T	0.2364	0.2362		0.2366	0.2362	7 T	0.2366	0.2364	7 T	0.2367	0.2364	8 T
7	0.2756	0.2753	0.2759	0.2755		0.2759	0.2756		0.2760	0.2756		0.2761	0.2758		0.2762	0.2758	
8	0.3150	0.3147	0.3153	0.3149	1 L	0.3153	0.3150		0.3154	0.3150	0 T	0.3155	0.3152	2 T	0.3156	0.3152	2 T
9	0.3543	0.3540	0.3546	0.3542	6 T	0.3546	0.3543		0.3547	0.3543	7 T	0.3548	0.3545	8 T	0.3549	0.3545	9 T
10	0.3937	0.3934	0.3940	0.3936		0.3940	0.3937					0.3939			0.3943	0.3939	
12	0.4724	0.4721	0.4727	0.4723		0.4728	0.4724					0.4727			0.4731	0.4727	
15	0.5906	0.5903	0.5909	0.5905	1 L	0.5910	0.5906					0.5909		3 T	0.5913	0.5909	3 T
17	0.6693	0.6690	0.6696	0.6692	6 T	0.6697	0.6693		0.6698	0.6693	8 T	0.6699	0.6696	9 T	0.6700	0.6696	10 T
20	0.7874	0.7870	0.7878	0.7872		0.7878	0.7875		0.7880	0.7875		0.7881	0.7877		0.7882	0.7877	
25	0.9843	0.9839	0.9847	0.9841	2 L	0.9847	0.9844		0.9849	0.9844	1 T	0.9850	0.9846	3 T	0.9851	0.9846	3 T
30	1.1811	1.1807	1.1815	1.1809	8 T	1.1815	1.1812		1.1817	1.1812	10 T	1.1818	1.1814	11 T	1.1819	1.1814	12 T
35	1.3780	1.3775	1.3784	1.3778		1.3785	1.3781		1.3787	1.3781		1.3788	1.3784		1.3790	1.3784	
40	1.5748	1.5743	1.5752	1.5746	2 L	1.5753	1.5749		1.5755	1.5749	1 T	1.5756	1.5752	4 T	1.5758	1.5752	4 T
45	1.7717	1.7712	1.7721	1.7715	9 T	1.7722	1.7718		1.7724	1.7718	12 T	1.7725	1.7721	13 T	1.7727	1.7721	15 T
50	1.9685	1.9680	1.9689	1.9683		1.9690	1.9686		1.9692	1.9686		1.9693	1.9689		1.9695	1.9689	
55	2.1654	2.1648	2.1658	2.1651		2.1660	2.1655		2.1662	2.1655		2.1663	2.1658		2.1666	2.1658	
60	2.3622	2.3616	2.3626	2.3619		2.3628	2.3623		2.3630	2.3623		2.3631	2.3626		2.3634	2.3626	
65	2.5591	2.5585	2.5595	2.5588	3 L	2.5597	2.5592		2.5599	2.5592	1 T	2.5600	2.5595	4 T	2.5603	2.5595	4 T
70	2.7559	2.7553	2.7563	2.7556	11 T	2.7565	2.7560		2.7567	2.7560	14 T	2.7568	2.7563	15 T	2.7571	2.7563	18 T
75	2.9528	2.9522	2.9532	2.9525		2.9534	2.9529		2.9536	2.9529		2.9537	2.9532		2.9540	2.9532	
80	3.1496	3.1490	3.1500	3.1493		3.1502	3.1497		3.1504	3.1497		3.1505	3.1500		3.1508	3.1500	
85	3.3465	3.3457	3.3470	3.3461		3.3472	3.3466		3.3475	3.3466		3.3476	3.3470		3.3479	3.3470	
90	3.5433	3.5425	3.5438	3.5429		3.5440	3.5434		3.5443	3.5434		3.5444	3.5438		3.5447	3.5438	
95	3.7402	3.7394	3.7407	3.7398		3.7409	3.7403		3.7412	3.7403		3.7413	3.7407		3.7416	3.7407	
100	3.9370	3.9362	3.9375	3.9366		3.9377	3.9371		3.9380	3.9371		3.9381	3.9375		3.9384	3.9375	
105	4.1339	4.1331	4.1344	4.1335		4.1346	4.1340		4.1349	4.1340		4.1350	4.1344		4.1353	4.1344	
110	4.3307	4.3299	4.3312	4.3303	4 L	4.3314	4.3308		4.3317	4.3308	1 T	4.3318	4.3312	5 T	4.3321	4.3312	5 T
115	4.5276	4.5268	4.5281	4.5272	13 T	4.5283	4.5277		4.5286	4.5277	18 T	4.5287	4.5281	19 T	4.5290	4.5281	22 T
120	4.7244	4.7236	4.7249	4.7240		4.7251	4.7245		4.7254	4.7245		4.7255	4.7249		4.7258	4.7249	

Resultant fit

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Shaft and housing fits: resultant fit

$$k5 = 2.3628 - 2.3623$$

$$\text{bearing} = \frac{2.3616 - 2.3622}{.0012T - .0001T}$$

T = tight / L = loose

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Shaft and housing fits: selection of tolerance

Bearing number: NU212 ECM C3

Application: electric motor

Rotation: inner ring

Bearing type: cylindrical roller

Bore size: 60 mm

Tolerance symbol: ???

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Selection of tolerance...

Selection of Solid Steel Shaft Tolerance
Classification for Metric Radial Ball and Roller Bearings of Tolerance
Classes ABEC-1, RBEC-1
(Except Inch Dimensioned Tapered Roller Bearings)

Conditions	Shaft diameter, mm Ball bearings ¹	Cylindrical roller bearings, metric taper	Spherical roller bearings	Tolerance Symbol
Rotating inner ring load or direction of loading indeterminate				
Light loads	18≤100 100≤140	≤40 40≤100	— —	j6 k6
Normal loads	≤18 18≤100 100≤140 140≤200 200≤280 — — —	— ≤40 40≤100 100≤140 140≤200 200≤400 — — —	— ≤40 40≤100 65≤100 100≤140 140≤280 280≤500 >500	j5 k5 (k6) ² m5 (m6) ³ m6 n6 p6 r6 r7
Heavy loads	— — —	50≤140 140≤200 ≤200	50≤100 100≤140 >140	n6 ³ p6 ³ r6 ³
High demands on running accuracy with light loads	≤18 18≤100 100≤200 —	— ≤40 40≤140 140≤200	— — — —	h5 ⁴ j5 ⁴ k5 ⁴ m5 ⁴

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Selection of tolerance

Shaft Bearing-Seal Diameters (Values in inches) (cont.)

Bearing Diameters (mm)	Bore (inches)		j6		k5		k6		m5		m6	
	Max.	Min.	Shaft Dia. Max.	Fit in 0.0000"	Shaft Dia. Max.	Fit in 0.0000"	Shaft Dia. Max.	Fit in 0.0000"	Shaft Dia. Max.	Fit in 0.0000"	Shaft Dia. Max.	Fit in 0.0000"
4	0.1575	0.1572	0.1577	0.1574	0.1577	0.1575	0.1579	0.1575	0.1579	0.1577	0.1580	0.1577
5	0.1969	0.1966	0.1971	0.1968	0.1971	0.1969	0.1973	0.1969	0.1973	0.1971	0.1974	0.0001
6	0.2362	0.2359	0.2364	0.2361	0.2364	0.2362	0.2366	0.2362	0.2366	0.2364	0.2367	0.2364
7	0.2756	0.2753	0.2759	0.2755	0.2759	0.2756	0.2760	0.2756	0.2761	0.2758	0.2762	0.2758
8	0.3150	0.3147	0.3153	0.3149	0.3153	0.3150	0.3154	0.3150	0.3155	0.3152	0.3156	0.3152
9	0.3543	0.3540	0.3546	0.3542	0.3546	0.3543	0.3549	0.3545	0.3550	0.3545	0.3549	0.3545
10	0.3937	0.3934	0.3940	0.3936	0.3940	0.3937	0.3944	0.3939	0.3949	0.3939	0.3943	0.3939
12	0.4724	0.4721	0.4727	0.4723	0.4727	0.4724	0.4730	0.4727	0.4730	0.4727	0.4731	0.4727
15	0.5906	0.5903	0.5909	0.5905	0.5910	0.5906	0.5911	0.5906	0.5912	0.5909	0.5913	0.5909
17	0.6693	0.6690	0.6696	0.6692	0.6697	0.6693	0.6698	0.6693	0.6699	0.6696	0.6700	0.6696
20	0.7874	0.7870	0.7878	0.7872	0.7878	0.7875	0.7880	0.7875	0.7881	0.7877	0.7882	0.7877
25	0.9843	0.9839	0.9847	0.9841	0.9847	0.9844	0.9849	0.9844	0.9850	0.9846	0.9851	0.9846
30	1.1811	1.1807	1.1815	1.1809	1.1815	1.1812	1.1817	1.1812	1.1818	1.1814	1.1819	1.1814
35	1.3780	1.3775	1.3784	1.3778	1.3785	1.3781	1.3787	1.3781	1.3788	1.3784	1.3790	1.3784
40	1.5748	1.5743	1.5752	1.5746	1.5753	1.5749	1.5755	1.5749	1.5756	1.5752	1.5758	1.5752
45	1.7717	1.7712	1.7721	1.7715	1.7722	1.7718	1.7724	1.7718	1.7725	1.7721	1.7727	1.7721
50	1.9685	1.9680	1.9689	1.9683	1.9690	1.9686	1.9692	1.9686	1.9693	1.9689	1.9695	1.9689
55	2.1654	2.1648	2.1658	2.1651	2.1660	2.1655	2.1662	2.1655	2.1660	2.1658	2.1666	2.1658
60	2.3623	2.3617	2.3626	2.3619	2.3629	2.3623	2.3630	2.3623	2.3631	2.3626	2.3634	2.3626
65	2.5591	2.5585	2.5595	2.5588	2.5597	2.5592	2.5599	2.5592	2.5600	2.5595	2.5603	2.5595
70	2.7559	2.7553	2.7563	2.7556	2.7565	2.7560	2.7567	2.7560	2.7568	2.7563	2.7571	2.7563
75	2.9528	2.9522	2.9532	2.9525	2.9534	2.9529	2.9536	2.9529	2.9537	2.9532	2.9540	2.9532
80	3.1496	3.1490	3.1500	3.1493	3.1504	3.1497	3.1504	3.1497	3.1505	3.1500	3.1508	3.1500
85	3.3465	3.3459	3.3469	3.3462	3.3470	3.3464	3.3475	3.3466	3.3476	3.3470	3.3479	3.3470
90	3.5433	3.5427	3.5437	3.5430	3.5440	3.5434	3.5443	3.5434	3.5444	3.5438	3.5447	3.5438
95	3.7402	3.7396	3.7407	3.7398	3.7409	3.7403	3.7412	3.7403	3.7413	3.7407	3.7416	3.7407
100	3.9370	3.9362	3.9375	3.9366	3.9377	3.9371	3.9380	3.9371	3.9381	3.9375	3.9384	3.9375
105	4.1339	4.1331	4.1344	4.1335	4.1346	4.1340	4.1349	4.1340	4.1350	4.1344	4.1353	4.1344
110	4.3307	4.3299	4.3312	4.3303	4.3314	4.3308	4.3317	4.3308	4.3318	4.3312	4.3321	4.3312
115	4.5276	4.5268	4.5281	4.5272	4.5283	4.5277	4.5286	4.5277	4.5287	4.5281	4.5290	4.5281
120	4.7244	4.7236	4.7249	4.7240	4.7251	4.7245	4.7254	4.7245	4.7255	4.7249	4.7258	4.7249

Match the bearing bore

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Shaft and housing fits: selection of tolerance

2.3631 – 2.3626

2.3631

2.3626

.0005

Out of roundness and taper should be half the tolerance

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Selection of tolerance

Shaft Bearing-Seal Diameters (Values in inches) (cont.)

Bearing Diameters (mm)	Bore (inches)		j6		Fit in 0.0000"	k5		Fit in 0.0000"	k6		Fit in 0.0000"	m5		Fit in 0.0000"	m6		Fit in 0.0000"
	Max.	Min.	Max.	Min.		Max.	Min.		Max.	Min.		Max.	Min.		Max.	Min.	
4	0.1575	0.1572	0.1577	0.1574		0.1577	0.1575		0.1579	0.1575		0.1579	0.1577		0.1580	0.1577	
5	0.1969	0.1966	0.1971	0.1968	1 L	0.1971	0.1969	0 T	0.1973	0.1969	0 T	0.1973	0.1971	2 T	0.1974	0.0001	2 T
6	0.2362	0.2359	0.2364	0.2361	5 T	0.2364	0.2362	5 T	0.2366	0.2362	7 T	0.2366	0.2364	7 T	0.2367	0.2364	8 T
7	0.2756	0.2753	0.2759	0.2755		0.2759	0.2756		0.2760	0.2756		0.2761	0.2758		0.2762	0.2758	
8	0.3150	0.3147	0.3153	0.3149	1 L	0.3153	0.3150	0 T	0.3154	0.3150	0 T	0.3155	0.3152	2 T	0.3156	0.3152	2 T
9	0.3543	0.3540	0.3546	0.3542	6 T	0.3546	0.3543	6 T	0.3547	0.3543	7 T	0.3549	0.3545	7 T	0.3549	0.3545	8 T
10	0.3937	0.3934	0.3940	0.3936	6 T	0.3940	0.3937	6 T	0.3941	0.3937	7 T	0.3942	0.3939	7 T	0.3943	0.3939	9 T
12	0.4724	0.4721	0.4727	0.4723		0.4728	0.4724		0.4729	0.4724		0.4730	0.4727		0.4731	0.4727	
15	0.5906	0.5903	0.5909	0.5905	1 L	0.5910	0.5906	0 T	0.5911	0.5906	0 T	0.5912	0.5909	3 T	0.5913	0.5909	3 T
17	0.6693	0.6690	0.6696	0.6692	6 T	0.6697	0.6693	7 T	0.6698	0.6693	8 T	0.6699	0.6696	9 T	0.6700	0.6696	10 T
20	0.7874	0.7870	0.7878	0.7872		0.7878	0.7875		0.7880	0.7875		0.7881	0.7877		0.7882	0.7877	
25	0.9843	0.9839	0.9847	0.9841	2 L	0.9847	0.9844	1 T	0.9849	0.9844	1 T	0.9850	0.9846	3 T	0.9851	0.9846	3 T
30	1.1811	1.1807	1.1815	1.1809	8 T	1.1815	1.1812	8 T	1.1817	1.1812	10 T	1.1818	1.1814	11 T	1.1819	1.1814	12 T
35	1.3780	1.3775	1.3784	1.3778		1.3785	1.3781		1.3787	1.3781		1.3788	1.3784		1.3790	1.3784	
40	1.5748	1.5743	1.5752	1.5746	2 L	1.5753	1.5749	1 T	1.5755	1.5749	1 T	1.5756	1.5752	4 T	1.5758	1.5752	4 T
45	1.7717	1.7712	1.7721	1.7715	9 T	1.7722	1.7718	10 T	1.7724	1.7718	12 T	1.7725	1.7721	13 T	1.7727	1.7721	15 T
50	1.9685	1.9680	1.9689	1.9683		1.9690	1.9686		1.9692	1.9686		1.9693	1.9689		1.9695	1.9689	
55	2.1654	2.1648	2.1658	2.1651		2.1660	2.1655		2.1662	2.1655		2.1663	2.1658		2.1666	2.1658	
60	2.3622	2.3616	2.3626	2.3619		2.3628	2.3623		2.3630	2.3623		2.3631	2.3626		2.3634	2.3626	
65	2.5591	2.5585	2.5595	2.5588	3 L	2.5597	2.5592		2.5600	2.5595		2.5600	2.5595		2.5603	2.5595	4 T
70	2.7559	2.7553	2.7563	2.7556	11 T	2.7565	2.7560		2.7568	2.7563		2.7568	2.7563		2.7571	2.7563	18 T
75	2.9528	2.9522	2.9532	2.9525		2.9534	2.9529		2.9537	2.9532		2.9537	2.9532		2.9540	2.9532	
80	3.1496	3.1490	3.1500	3.1493		3.1502	3.1497		3.1505	3.1500		3.1505	3.1500		3.1508	3.1500	
85	3.3465	3.3457	3.3470	3.3461		3.3472	3.3466		3.3475	3.3466		3.3476	3.3470		3.3479	3.3470	
90	3.5433	3.5425	3.5438	3.5429		3.5440	3.5434		3.5443	3.5434		3.5444	3.5438		3.5447	3.5438	
95	3.7402	3.7394	3.7407	3.7398		3.7409	3.7403		3.7412	3.7403		3.7413	3.7407		3.7416	3.7407	
100	3.9370	3.9362	3.9375	3.9366	4 L	3.9377	3.9371	1 T	3.9380	3.9371	1 T	3.9381	3.9375	5 T	3.9384	3.9375	5 T
105	4.1339	4.1331	4.1344	4.1335	13 T	4.1346	4.1340	15 T	4.1349	4.1340	18 T	4.1350	4.1344	19 T	4.1353	4.1344	22 T
110	4.3307	4.3299	4.3312	4.3303		4.3314	4.3308		4.3317	4.3308		4.3318	4.3312		4.3321	4.3312	
115	4.5276	4.5268	4.5281	4.5272		4.5283	4.5277		4.5286	4.5277		4.5287	4.5281		4.5290	4.5281	
120	4.7244	4.7236	4.7249	4.7240		4.7251	4.7245		4.7254	4.7245		4.7255	4.7249		4.7258	4.7249	

Resultant fit

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Shaft and housing fits: resultant fit

$$m5 = 2.3631 - 2.3626$$

$$\text{bearing} = \frac{2.3616 - 2.3622}{.0015T - .0004T}$$

$$.0015T - .0004T$$

$$T = \text{tight} / L = \text{loose}$$

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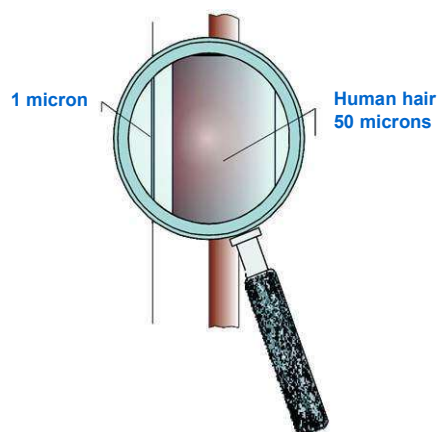
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Shaft and housing fits: relative sizes

Relative sizes:

- Paper clip: ~ 0.03"
- Human hair: ~ 0.002"
- Dirt particle: ~ 0.0002"
- Clearance 6310/C3: ~ 0.0012"
- Oil film: ~ 0.00002"



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Shaft and housing fits: outside measurements

- Accurate tools for accurate results
- Capable and calibrated
- Requirement: repeatedly measure within $\pm 0.0002''$



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Selection of housing tolerance...

Selection of Steel or Cast Iron Housing
Tolerance Classification for
Metric *Radial* Ball and Roller Bearings
of Tolerance Classes ABEC-1, RBEC-1
(Except Inch Dimensioned Tapered Roller Bearings)

Conditions	Tolerance Symbol	Displacement of outer ring
SOLID HOUSINGS		
Rotating outer ring load		
Heavy loads on bearings in thin-walled housings, heavy shock loads	P7	Cannot be displaced
Normal loads and heavy loads	N7	Cannot be displaced
Light and variable loads	M7	Cannot be displaced
Direction of load indeterminate		
Heavy shock loads	M7	Cannot be displaced
Normal loads and heavy loads axial displacement of outer ring unnecessary	K7	Cannot be displaced as a rule
Accurate or silent running		
	K6'	Cannot be displaced as a rule
	H6 ²	Can be displaced
	H6	Can easily be displaced

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Selection of housing tolerance

Housing Bearing-Seat Diameters (Values in Inches)

Bearing Outside Diameter	F7				G7				H8				H7				H6			
	Inches		Housing Bore		Fit in 0.0000"	Housing Bore		Fit in 0.0000"	Housing Bore		Fit in 0.0000"	Housing Bore		Fit in 0.0000"	Housing Bore		Fit in 0.0000"			
	Max.	Min.	Min.	Max.		Min.	Max.		Min.	Max.		Min.	Max.		Min.	Max.				
mm	Max.	Min.	Min.	Max.	0.0000"	Min.	Max.	0.0000"	Min.	Max.	0.0000"	Min.	Max.	0.0000"	Min.	Max.	0.0000"			
16	0.6299	0.6296	0.6305	0.6312	18 L 6 L	0.6301	0.6308	12 L 2 L	0.6299	0.6310	14 L 0	0.6299	0.6306	10 L 0	0.6299	0.6303	7 L 0			
35	1.3780	1.3776	1.3790	1.4000		1.3784	1.3793		1.3780	1.3795	14 L 0	1.3780	1.3790		1.3780	1.3786	10 L 0			
37	1.4567	1.4563	1.4577	1.4587	24 L	1.4571	1.4580	17 L	1.4567	1.4582		1.4567	1.4577		1.4567	1.4573	10 L 0			
40	1.5748	1.5744	1.5758	1.5768	10 L	1.5752									1.5748	1.5754	7 L 0			
42	1.6535	1.6531	1.6545	1.6555		1.6539									1.6535	1.6541				
47	1.8504	1.8500	1.8514	1.8524		1.8508									1.8504	1.8510				
52	2.0472	2.0467	2.0484	2.0496		2.0476	2.0490		2.0472	2.0490		2.0472	2.0484		2.0472	2.0479				
55	2.1654	2.1649	2.1666	2.1678	29 L	2.1658	2.1670	21 L	2.1654	2.1672	23 L	2.1654	2.1666	17 L	2.1654	2.1661	12 L			
62	2.4409	2.4404	2.4421	2.4433		2.4413	2.4425		2.4409	2.4427		2.4409	2.4421		2.4409	2.4416				
68	2.6772	2.6767	2.6784	2.6796		2.6776	2.6788		2.6772	2.6790		2.6772	2.6784		2.6772	2.6779				
72	2.8346	2.8341	2.8358	2.8370	12 L	2.8350	2.8362	4 L	2.8346	2.8364	0	2.8346	2.8358	0	2.8346	2.8353	0			
75	2.9527	2.9				2.9527	2.9545		2.9527	2.9545		2.9527	2.9539		2.9527	2.9534				
80	3.1496	3.1				3.1496	3.1514		3.1496	3.1514		3.1496	3.1508		3.1496	3.1503				
85	3.3465	3.3459	3.3479	3.3493		3.3470	3.3484		3.3465	3.3486		3.3465	3.3479		3.3465	3.3474				
90	3.5433	3.5427	3.5447	3.5461		3.5438	3.5452		3.5433	3.5454		3.5433	3.5447		3.5433	3.5442				
95	3.7402	3.7396	3.7416	3.7430		3.7407	3.7421		3.7402	3.7423		3.7402	3.7416		3.7402	3.7411				
100	3.9370	3.9364	3.9384	3.9398	34 L	3.9375	3.9389	25 L	3.9370	3.9391	27 L	3.9370	3.9384	20 L	3.9370	3.9379	15 L			
110	4.3307	4.3301	4.3321	4.3335	11 L	4.3312	4.3326	6 L	4.3307	4.3328	0	4.3307	4.3321	0	4.3307	4.3316	0			
115	4.5276	4.5270	4.5290	4.5304		4.5281	4.5295		4.5276	4.5297		4.5276	4.5290		4.5276	4.5285				
120	4.7244	4.7238	4.7258	4.7272		4.7249	4.7263		4.7244	4.7265		4.7244	4.7258		4.7244	4.7253				
125	4.9213	4.9206	4.9220	4.9246		4.9219	4.9234		4.9213	4.9238		4.9213	4.9229		4.9213	4.9223				
130	5.1181	5.1174	5.1198	5.1214	40 L	5.1187	5.1202	28 L	5.1181	5.1206	32 L	5.1181	5.1197	23 L	5.1181	5.1191	17 L			
140	5.5118	5.5111	5.5135	5.5151	17 L	5.5124	5.5139	6 L	5.5118	5.5143		5.5118	5.5134	0	5.5118	5.5128	0			
145	5.7087	5.7080	5.7104	5.7120		5.7093	5.7108		5.7087	5.7112		5.7087	5.7103		5.7087	5.7097				
150	5.9055	5.9048	5.9072	5.9088		5.9061	5.9076		5.9055	5.9080		5.9055	5.9071		5.9055	5.9065				
160	6.2992	6.2982	6.3009	6.3025		6.2998	6.3013		6.2992	6.3017		6.2992	6.3008		6.2992	6.3002				
165	6.4961	6.4951	6.4978	6.4994	43 L	6.4967	6.4982	31 L	6.4961	6.4986	35 L	6.4961	6.4977		6.4961	6.4971				
170	6.6929	6.6919	6.6946	6.6962		6.6935	6.6950		6.6929	6.6954		6.6929	6.6944		6.6929	6.6939				
180	7.0866	7.0856	7.0883	7.0899	17 L	7.0872	7.0887	6 L	7.0866	7.0891		7.0866	7.0881		7.0866	7.0876				

With the tolerance symbol

Match the bearing OD

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Shaft and housing fits: selection of tolerance

4.3307 – 4.3316

4.3307

4.3316

.0009

Out of roundness and taper should be half the tolerance

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Selection of housing tolerance

Housing Bearing-Seat Diameters (Values in Inches)																							
Bearing Outside Diameter				F7				G7				H8				H7				H6			
mm	Inches		Housing Bore		Fit in housing bore	Housing Bore		Fit in housing bore	Housing Bore		Fit in housing bore	Housing Bore		Fit in housing bore	Housing Bore		Fit in housing bore	Housing Bore		Fit in housing bore			
	Max.	Min.	Min.	Max.		Min.	Max.		Min.	Max.		Min.	Max.		Min.	Max.		Min.	Max.		Min.	Max.	
16	0.6299	0.6296	0.6305	0.6312	0.0000	0.6301	0.6308	0.0000	0.6299	0.6310	0.0000	0.6299	0.6306	0.0000	0.6299	0.6307	0.0000	0.6299	0.6307	0.0000			
32	1.2598	1.2594	1.2608	1.2618	0.0000	1.2602	1.2611	0.0000	1.2598	1.2613	0.0000	1.2598	1.2608	0.0000	1.2598	1.2604	0.0000	1.2598	1.2604	0.0000			
35	1.3780	1.3776	1.3790	1.4000	0.0000	1.3784	1.3793	0.0000	1.3780	1.3795	0.0000	1.3780	1.3790	0.0000	1.3780	1.3786	0.0000	1.3780	1.3786	0.0000			
37	1.4567	1.4563	1.4577	1.4587	0.0000	1.4571	1.4580	0.0000	1.4567	1.4582	0.0000	1.4567	1.4577	0.0000	1.4567	1.4573	0.0000	1.4567	1.4573	0.0000			
40	1.5748	1.5744	1.5758	1.5768	0.0000	1.5752	1.5761	0.0000	1.5748	1.5763	0.0000	1.5748	1.5758	0.0000	1.5748	1.5754	0.0000	1.5748	1.5754	0.0000			
42	1.6535	1.6531	1.6545	1.6555	0.0000	1.6539	1.6548	0.0000	1.6535	1.6550	0.0000	1.6535	1.6545	0.0000	1.6535	1.6541	0.0000	1.6535	1.6541	0.0000			
47	1.8504	1.8500	1.8514	1.8524	0.0000	1.8508	1.8517	0.0000	1.8504	1.8519	0.0000	1.8504	1.8514	0.0000	1.8504	1.8510	0.0000	1.8504	1.8510	0.0000			
52	2.0472	2.0467	2.0484	2.0496	0.0000	2.0476	2.0488	0.0000	2.0472	2.0490	0.0000	2.0472	2.0484	0.0000	2.0472	2.0479	0.0000	2.0472	2.0479	0.0000			
55	2.1654	2.1649	2.1666	2.1678	0.0000	2.1658	2.1670	0.0000	2.1654	2.1672	0.0000	2.1654	2.1666	0.0000	2.1654	2.1661	0.0000	2.1654	2.1661	0.0000			
62	2.4409	2.4404	2.4421	2.4433	0.0000	2.4413	2.4425	0.0000	2.4409	2.4427	0.0000	2.4409	2.4421	0.0000	2.4409	2.4416	0.0000	2.4409	2.4416	0.0000			
68	2.6772	2.6767	2.6784	2.6796	0.0000	2.6776	2.6788	0.0000	2.6772	2.6790	0.0000	2.6772	2.6784	0.0000	2.6772	2.6779	0.0000	2.6772	2.6779	0.0000			
72	2.8346	2.8341	2.8358	2.8370	0.0000	2.8350	2.8362	0.0000	2.8346	2.8364	0.0000	2.8346	2.8358	0.0000	2.8346	2.8353	0.0000	2.8346	2.8353	0.0000			
75	2.9527	2.9522	2.9539	2.9551	0.0000	2.9532	2.9543	0.0000	2.9527	2.9545	0.0000	2.9527	2.9539	0.0000	2.9527	2.9534	0.0000	2.9527	2.9534	0.0000			
80	3.1496	3.1491	3.1508	3.1520	0.0000	3.1500	3.1512	0.0000	3.1496	3.1514	0.0000	3.1496	3.1508	0.0000	3.1496	3.1503	0.0000	3.1496	3.1503	0.0000			
85	3.3465	3.3459	3.3479	3.3493	0.0000	3.3470	3.3484	0.0000	3.3465	3.3484	0.0000	3.3465	3.3479	0.0000	3.3465	3.3474	0.0000	3.3465	3.3474	0.0000			
90	3.5433	3.5427	3.5447	3.5461	0.0000	3.5438	3.5452	0.0000	3.5433	3.5452	0.0000	3.5433	3.5447	0.0000	3.5433	3.5442	0.0000	3.5433	3.5442	0.0000			
95	3.7402	3.7396	3.7416	3.7430	0.0000	3.7407	3.7421	0.0000	3.7402	3.7421	0.0000	3.7402	3.7416	0.0000	3.7402	3.7411	0.0000	3.7402	3.7411	0.0000			
100	3.9370	3.9364	3.9384	3.9398	0.0000	3.9375	3.9389	0.0000	3.9370	3.9391	0.0000	3.9370	3.9384	0.0000	3.9370	3.9379	0.0000	3.9370	3.9379	0.0000			
110	4.3307	4.3301	4.3321	4.3335	0.0000	4.3312	4.3326	0.0000	4.3307	4.3329	0.0000	4.3307	4.3321	0.0000	4.3307	4.3316	0.0000	4.3307	4.3316	0.0000			
115	4.5276	4.5270	4.5290	4.5304	0.0000	4.5281	4.5295	0.0000	4.5276	4.5297	0.0000	4.5276	4.5290	0.0000	4.5276	4.5286	0.0000	4.5276	4.5286	0.0000			
120	4.7244	4.7238	4.7258	4.7272	0.0000	4.7249	4.7263	0.0000	4.7244	4.7265	0.0000	4.7244	4.7258	0.0000	4.7244	4.7253	0.0000	4.7244	4.7253	0.0000			
125	4.9213	4.9206	4.9226	4.9240	0.0000	4.9218	4.9232	0.0000	4.9213	4.9236	0.0000	4.9213	4.9229	0.0000	4.9213	4.9223	0.0000	4.9213	4.9223	0.0000			
130	5.1181	5.1174	5.1194	5.1214	0.0000	5.1186	5.1202	0.0000	5.1181	5.1206	0.0000	5.1181	5.1197	0.0000	5.1181	5.1191	0.0000	5.1181	5.1191	0.0000			
140	5.5118	5.5111	5.5131	5.5151	0.0000	5.5124	5.5139	0.0000	5.5118	5.5143	0.0000	5.5118	5.5134	0.0000	5.5118	5.5128	0.0000	5.5118	5.5128	0.0000			
145	5.7087	5.7080	5.7104	5.7120	0.0000	5.7093	5.7108	0.0000	5.7087	5.7112	0.0000	5.7087	5.7103	0.0000	5.7087	5.7097	0.0000	5.7087	5.7097	0.0000			
150	5.9055	5.9048	5.9072	5.9088	0.0000	5.9061	5.9076	0.0000	5.9055	5.9080	0.0000	5.9055	5.9071	0.0000	5.9055	5.9065	0.0000	5.9055	5.9065	0.0000			
160	6.2992	6.2982	6.3009	6.3025	0.0000	6.2998	6.3013	0.0000	6.2992	6.3017	0.0000	6.2992	6.3011	0.0000	6.2992	6.3002	0.0000	6.2992	6.3002	0.0000			
165	6.4961	6.4951	6.4978	6.4994	0.0000	6.4967	6.4982	0.0000	6.4961	6.4986	0.0000	6.4961	6.4980	0.0000	6.4961	6.4971	0.0000	6.4961	6.4971	0.0000			
170	6.6929	6.6919	6.6946	6.6962	0.0000	6.6935	6.6950	0.0000	6.6929	6.6954	0.0000	6.6929	6.6949	0.0000	6.6929	6.6939	0.0000	6.6929	6.6939	0.0000			
180	7.0866	7.0856	7.0883	7.0899	0.0000	7.0872	7.0887	0.0000	7.0866	7.0891	0.0000	7.0866	7.0886	0.0000	7.0866	7.0876	0.0000	7.0866	7.0876	0.0000			

Resultant fit

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Resultant fit

15 L

88

Shaft and housing fits: resultant fit

$$H6 = 4.3307 - 4.3316$$

$$\text{bearing} = 4.3307 - 4.3301$$

$$.0000 - .0015L$$

T = tight / L = loose

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Hands-on activity #2

Hands-on activity:

- Measuring for Conformance
- Follow the Hands-on Activity Sheet #2

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Video: shaft conformance measurement

- Video shows best practices for shaft conformance measurement
- Discuss: Do I really have to wear gloves?

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Shaft and housing fits: measurements

- Inside measurements are difficult to take without proper equipment
- A 3-point mike, or gauge:
 - Very good repeatability
 - Accuracy not user dependent
 - Self-centering
 - Self-aligning

Requirement: repeatedly within $\pm 0.0002''$

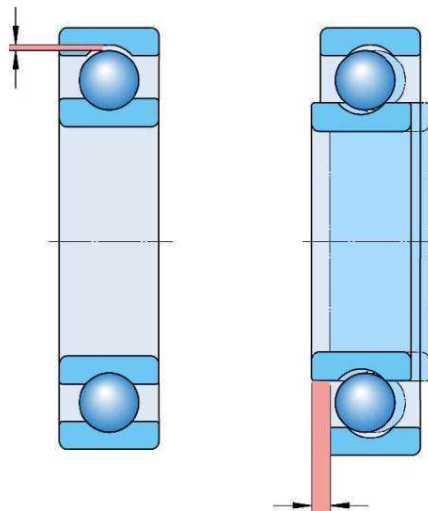


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Shaft and housing fits: radial & axial clearance...



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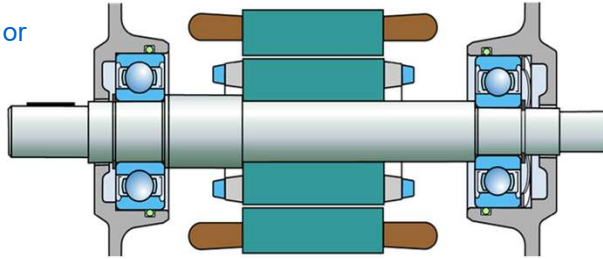
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Shaft and housing fits: radial & axial clearance

Aluminum housing

- A rule of thumb: use a tolerance one step tighter than for steel or iron housing
- E.g., if H7 for steel, use J7 for aluminum housing
- At loose fit, use an O-ring mounted in a groove in the bearing seating



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Shaft and housing fits: ineffective repairs

- Gluing the bearing to the shaft
- Side clamping
- Knurling/peening



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Shaft and housing fits: restoring fits

1. Replace the shaft
2. Emergency solutions – get approval from your customer
 - Sleeve the shaft or housing
 - *Electroplating*
 - *Metal Spraying*
 - *Emergency: welding and grinding*
3. These procedure repairs will require a high degree of craftsmanship and machining to original size

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Motor assembly and bearing mounting

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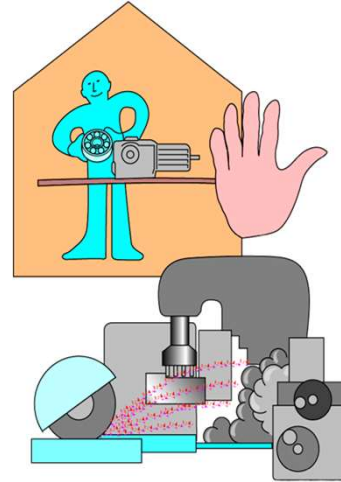
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Motor assembly: keep it clean!

- Arrange a clean working place
- Always wipe off workbench and tools
- Assemble in separate room or away from contaminating processes
- Avoid open airflow through the shop in assembly area
- Cover/protect work in progress from dust



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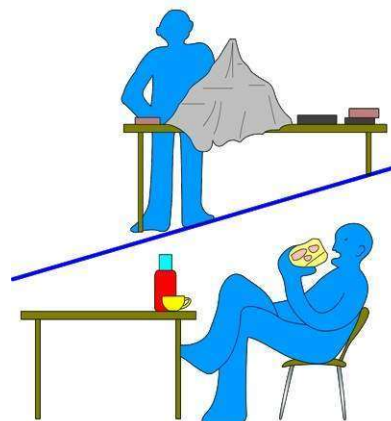
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Motor assembly: “work in progress”

Work in progress: All Motor Parts
Shaft, end bells, bearings, seals,
brushes, etc.

- Protect work in progress from damage, corrosion, dirt
- Avoid or eliminate dirt and dust from the shop environment
- Treat machined surfaces with anti-corrosive
- Store work in progress in safe, protected, and designated areas



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Clean working place



100

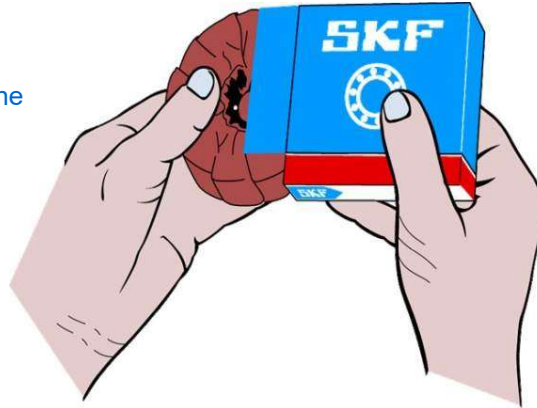
Bearing mounting: preparation

- Inspect bearing and check for damaged packaging
- If damaged, return for field scrap credit with your local SKF Distributor or contact SKF Customer Service
- Return the bearing if slushing compound is oxidized (dry/powdery)
- Don't clean new bearings

101

Bearing packaging

- Keep the bearing in the box as long as possible
- Do not remove the rust inhibiting compound on the bearing surfaces!



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Bearing storage

FIFO date code

Vibe absorbing material



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Mounting: exchange of bearings

The trend:

- Open deep groove ball bearings are replaced by sealed or shielded ones
 - Seals -2RS1
 - At high speed seals -2RZ
 - Shields -2Z
- Caution: Check grease life!



Advantages:

- Bearings and grease are kept clean
- No greasing – simplified mounting
- Longer grease and service life

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Mounting: online resources for maintenance

Start with: www.mapro.skf.com

- Cold and hot mounting
- SKF Oil Injection Method
- SKF Drive-Up Method

Mounting: www.skf.com/mount

SKF Support

The SKF Oil Injection CD-ROM
The CD-ROM publication series (PDFs), contains a calculation program, which also includes the calculation of mounting force and oil injection method. The CD-ROM provides detailed technical and practical information on how to use the SKF Oil Injection Method for mounting and dismounting bearings. It also provides information on the calculation of mounting force and oil injection method. The program also includes information on the calculation of mounting force and oil injection method. The program also includes information on the calculation of mounting force and oil injection method.

The SKF Drive-Up Method CD-ROM
The CD-ROM publication series (PDFs), is used to calculate the required pressure to use the SKF Drive-Up Method. It also includes the calculation of mounting force and oil injection method. The CD-ROM provides detailed technical and practical information on how to use the SKF Drive-Up Method for mounting and dismounting bearings. It also provides information on the calculation of mounting force and oil injection method. The program also includes information on the calculation of mounting force and oil injection method. The program also includes information on the calculation of mounting force and oil injection method.

Online Mounting and Dismounting Instructions
At several key locations, SKF offers mounting and dismounting instructions. These instructions are available online and are also available in PDF format. The instructions are available in several languages and are also available in PDF format. The instructions are available in several languages and are also available in PDF format.

SKF Maintenance and Lubrication Products Catalogue
The catalogue, published in several languages, contains detailed information and technical data on the various SKF mounting and dismounting products. The catalogue is available online and is also available in PDF format. The catalogue is available in several languages and is also available in PDF format.

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Mounting: which method?

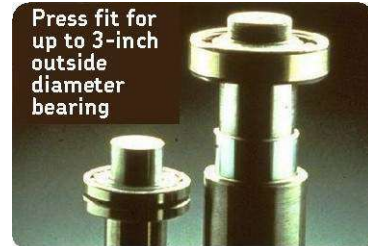
Cold or temperature mounting?

1. Cold mounting can be accomplished with:

- Impact drivers
- Nut and spanner
- Arbor presses
- Hydraulic assist

2. Temperature mounting

- Try to achieve minimum 150° F temperature difference between bearing and mating component



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Mounting: the right way and wrong way!



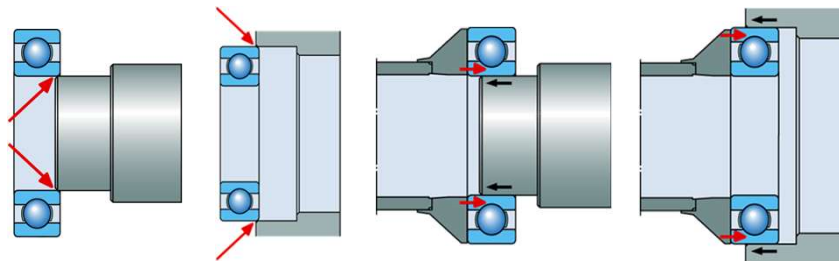
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Mounting: cold mounting



Interference fit on the shaft
or in the housing

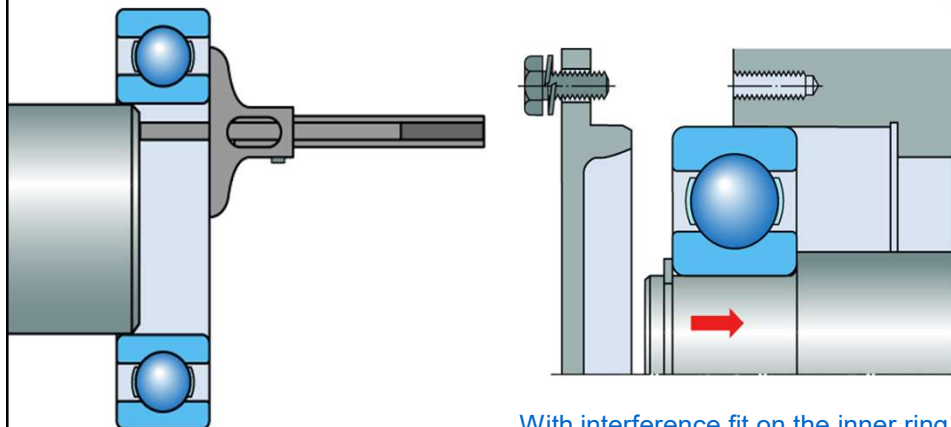
Always apply the pressure on the
ring with the interference fit

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Mounting: additional checks and notes...



Align the bearing

With interference fit on the inner ring,
mount the bearing on the shaft first

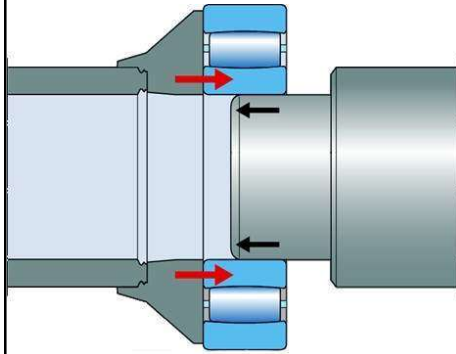
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Mounting: additional checks and notes



When mounting a CARB bearing on a shaft with interference fit, also support the outer ring



INSOCOAT and hybrid bearings are mounted in the same manner as the basic bearing type

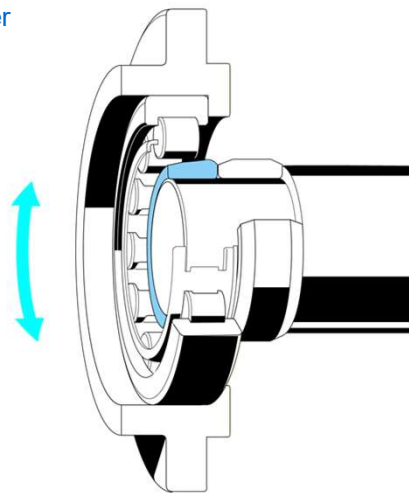
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Mounting: cylindrical roller bearings...

- When mounting a cylindrical roller bearing, lubricate the raceway and rollers
- Use an entering sleeve
- Rotate the assembly and fit carefully



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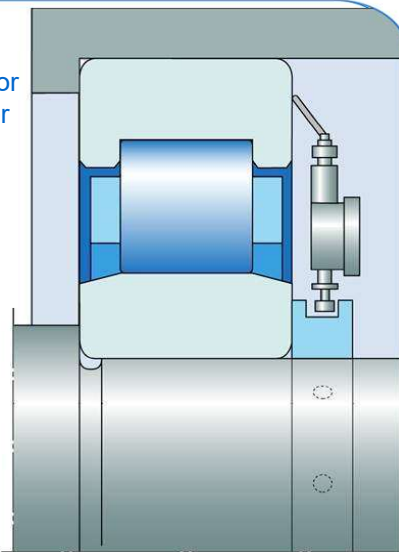
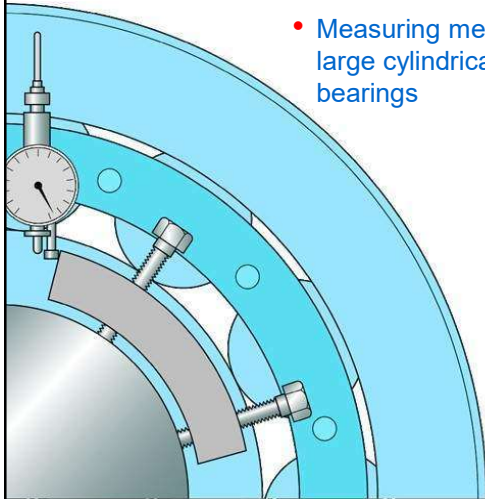
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Mounting: cylindrical roller bearings

Alignment check

- Measuring method for large cylindrical roller bearings



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Mounting: using heat



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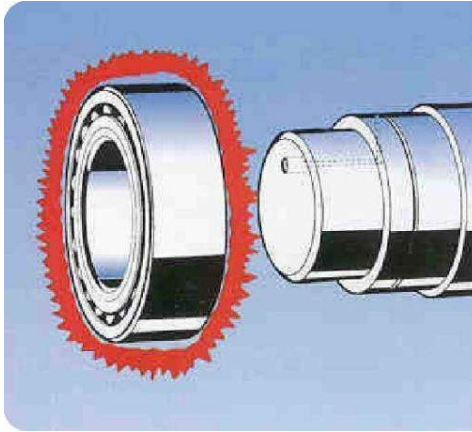
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Mounting: temperature...

- Used on bearings over 100 mm O.D. (over 50 mm shaft)
- Temperature rise of 150° F over ambient is sufficient
- Bearing rings will expand about 0.001" for every inch of shaft diameter



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Mounting: temperature...

Never heat a bearing above 250° F:

- Seals may burn or crack
- Slushing compound may oxidize
- Grease in capped bearings may separate or oxidize
- Use temperature monitoring

Hold the bearing against the shaft shoulder:

- Bearing may change position as heat transfer occurs



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Mounting: temperature

Electric Hot Plates

Advantages:

- Controlled temperature, clean (if covered)
- Best for large quantities of small bearings

Disadvantages:

- Long heating time

Ovens

Advantages:

- Temp. control, cleaner, uniform heating

Disadvantages:

- Dirty, costly, long heating time

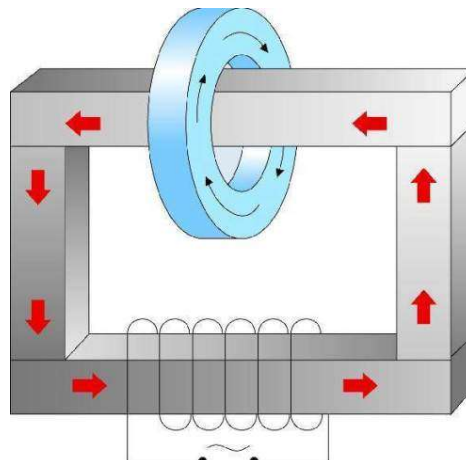
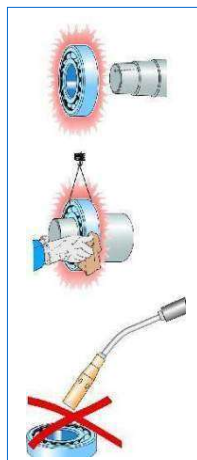


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Mounting: principles of induction heating



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Mounting: induction heating – heaters



Portable Induction Heater



Strength of the heater is chosen by the size of the bearings



Energy efficient, safe, and fast

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Mounting: induction heating and magnetism

- A magnetized bearing will attract dust and dirt!



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Separate your mounting area



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Mounting: oil bath

Hot Oil Bath**

Advantages:

- Uniform Heating

Disadvantages:

- Messy
- Bearing Contamination
- Long Heating Times
- Health Hazard
- Fire Hazard
- Disposal Problem



** Make Sure Oil Has a Flash Point Greater Than 250° C (480° F)

121

Mounting: hot mounting field experiences



122

Mounting: non-recommended heating

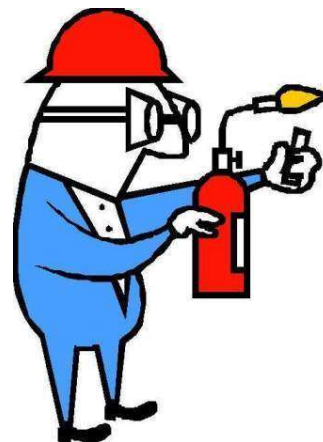
Torch

Advantages

- Bearing manufacturers sell more bearings
- Speed

Disadvantages

- Potential risk of damaging the bearing and shaft
- Non-uniform heating
- No temperature control
- Fire hazard
- Health hazard



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Mounting: cooling the shaft

Cooling can also be effective

- Refrigerator or freezer
- Ice (use a bag to prevent corrosion)
- Dry ice is permissible
- Liquid nitrogen is too cold
- Ensure components are dry before completing installation
- 150° F temperature difference is generally sufficient



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Motor assembly: summary...

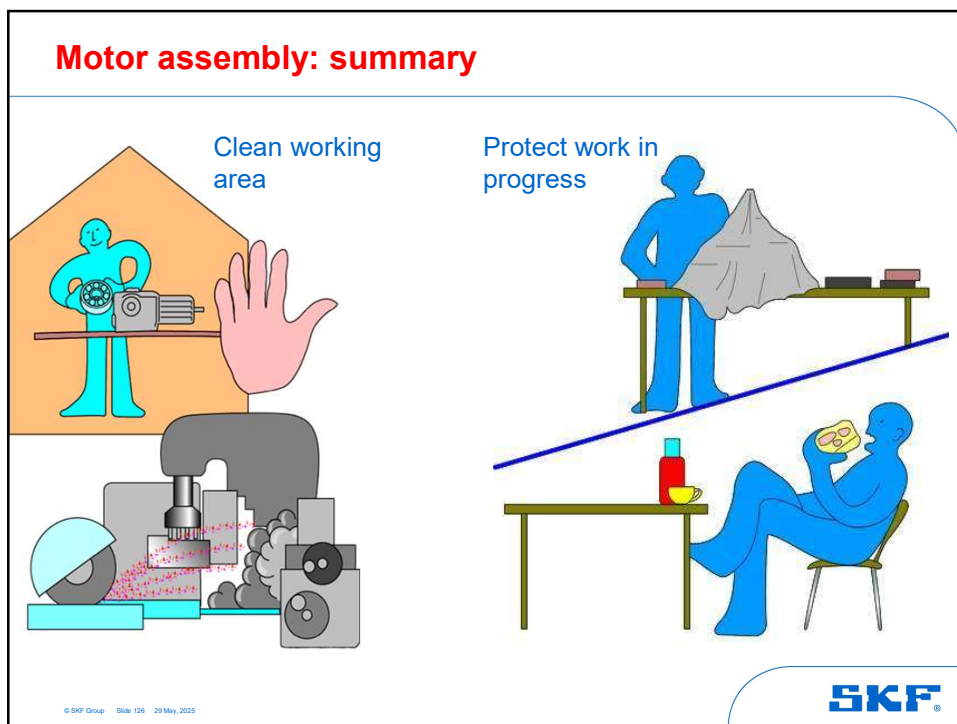
- Make sure to arrange a clean workplace
- Always check and document bearing fits before mounting
- Always use capable and calibrated gauges/micrometers
- Temperature methods need controls to avoid overheating
- Induction heating is a fast, safe, and reliable method
- Make sure your induction heater demagnetizes the bearings

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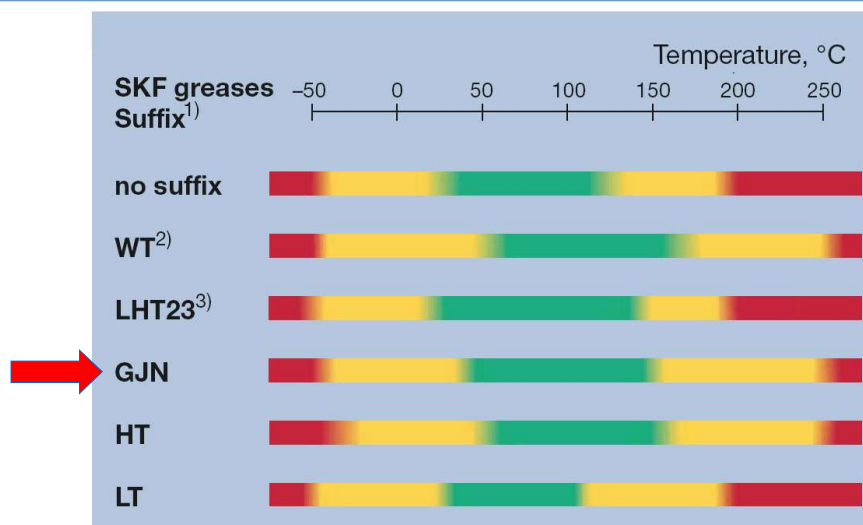


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Motor lubrication

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Lubrication: SKF factory fill EM greases



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Common error: mixing greases

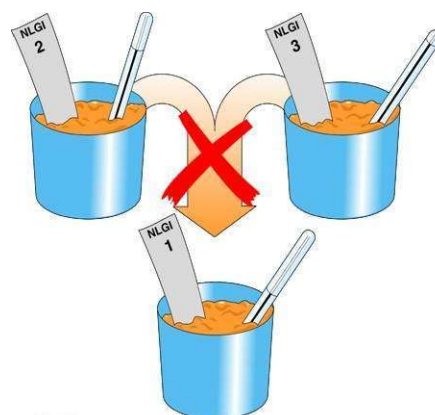
Commonly recommended Electric Motor Grease:

- **Polyurea** thickener

Commonly recommended grease for "balance of plant":

- **Lithium Complex** thickener

INCOMPATIBLE when mixed



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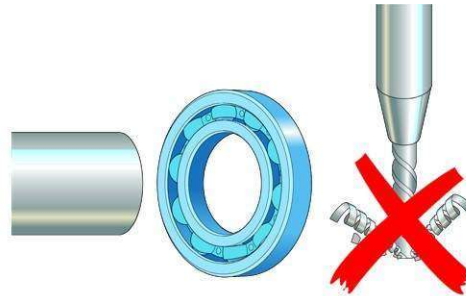
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Lubrication: greasing procedures



Keep the bearing in the box as long as possible



Clean and dust-free working place

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Lubrication: initial greasing procedures

Make sure to:

1. Use clean grease from a sealed grease container
2. Use the right quality of grease -no mixing!
3. Apply the grease immediately after mounting
4. Apply the correct quantity
 - Bearing filled to 100% (except CARB – 50%)
 - Housing filled to 30–50%
 - Never overfill – hot running may occur

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Markings and caps



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Grease tubes



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Lubrication: contamination



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Lubrication: real issues



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Lubrication: grease application

- Fill the free space in the housing with grease to 30–50%
- Clean and flush grease fitting and duct
- Check that the grease has penetrated through the bearing: 100% fill

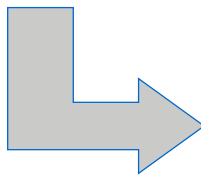


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Lubrication: typical lubrication mistakes



LAGG 1M -
1kg grease pump

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Lubrication: real issues



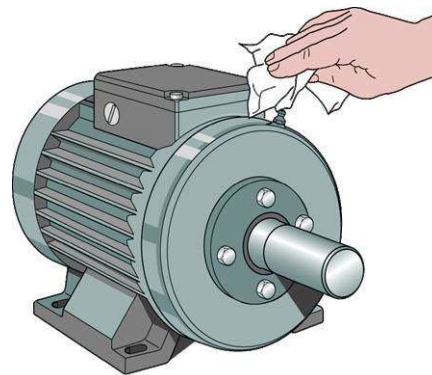
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Lubrication: grease ducts and fittings

- Grease ducts/tubes of black iron will corrode
- Seamless, stainless grease ducts are preferred
- Rust and dirt shorten bearing and motor service life
- Always clean out grease ducts
- Treat ducts with anti-corrosive
- Replace grease fittings when possible
- Flush with fresh grease



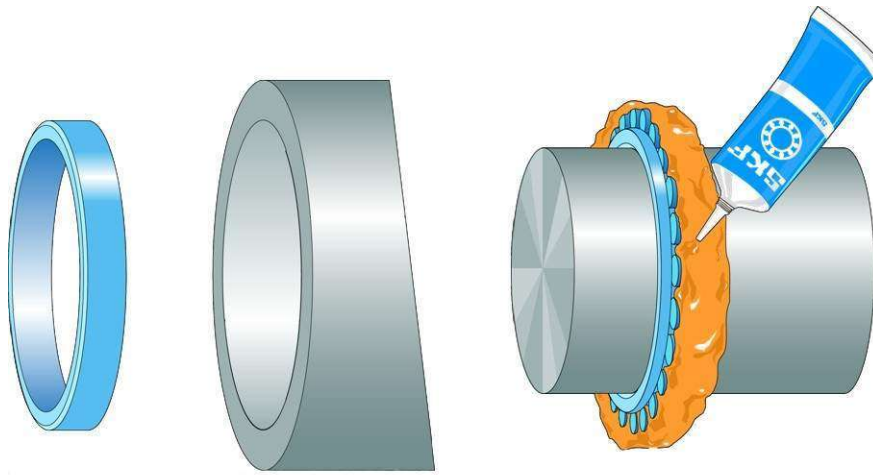
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Lubrication: greasing separable bearings



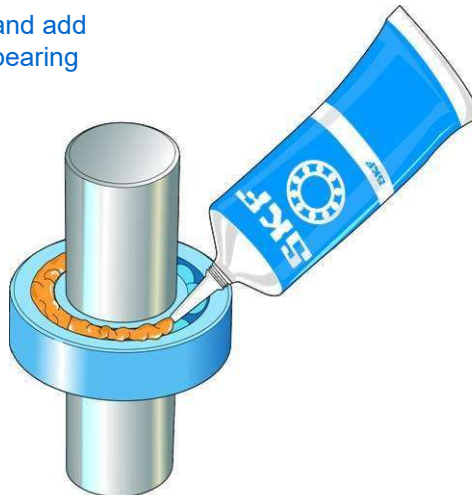
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Lubrication: vertical shafts

- Supply grease from above and add a reserve of grease on the bearing upper side



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Preventing over-greasing: use a SPAL

SPAL = Single
Point
Automatic
Lubricator

- Use SKF Dialset program for all grease re-lube calculations

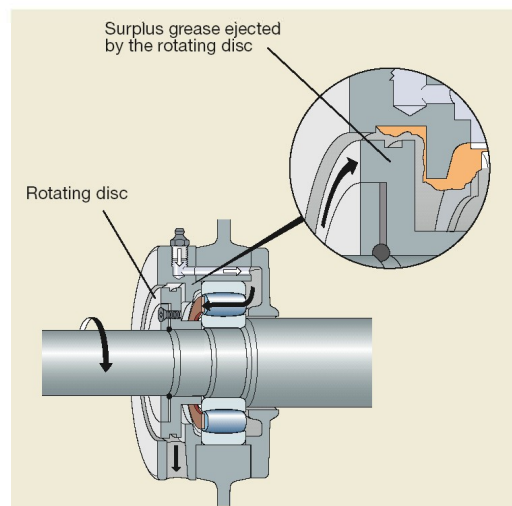


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Lubrication: grease escape valve



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Post motor service procedures

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Document every step of the repair process

- Incoming inspection
 - Document the “as found” condition
 - Estimate necessary repairs
- Document the “as returned” condition
 - Repairs and work performed
 - Parts used and replaced
 - Measurements taken
- Final conformance/acceptance tests
 - Electrical integrity
 - Mechanical integrity: vibration and balancing tests

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Every final test/conformance test should include:

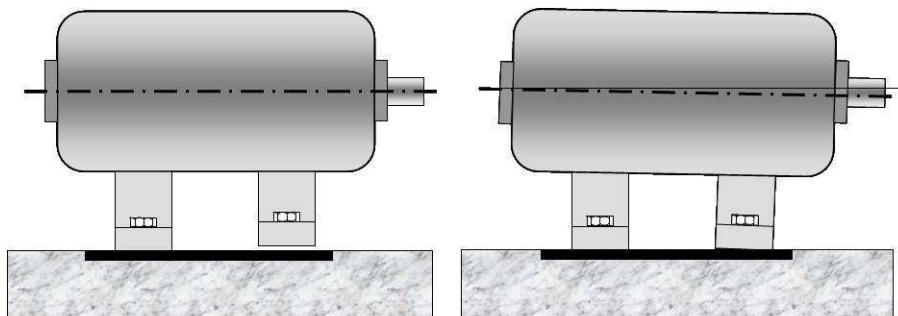
- Mechanical integrity: Check and document
 - Foot flatness
 - Vibration levels
 - Motor temperature
 - Rotor balancing
 - Shaft run out
- Check for circulating currents (DC motors)
- And any site-specific requirements

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Why check motor foot flatness?



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Check conformance: vibration levels

- Check conformance against vibration standards
- SKF Certification program has its own vibration standards
- Some customers have their own
- NEMA, ISO, API, other common standards
- Use a tool capable of measuring and analyzing velocity and acceleration readings



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ISO 10816-1 overall guidelines

Velocity Severity		Velocity Range Limits and Machine Classes ISO Standard 10816-1			
mm/s RMS	in/s PEAK	Small Machines Class I	Medium Machines Class II	Large Machines	
				Rigid Supports Class III	Less Rigid Supports Class IV
0.28	0.02	Good	Good	Good	Good
0.45	0.03				
0.71	0.04				
1.12	0.06	Satisfactory	Satisfactory	Satisfactory	Satisfactory
1.80	0.10	Unsatisfactory (Alert)			
2.80	0.16	Unacceptable (Danger)	Unsatisfactory (Alert)	Unsatisfactory (Alert)	Unsatisfactory (Alert)
4.50	0.25		Unacceptable (Danger)	Unacceptable (Danger)	Unacceptable (Danger)
7.10	0.40			Unacceptable (Danger)	Unacceptable (Danger)
11.20	0.62				
18.00	1.00				
28.00	1.56				
45.00	2.51				

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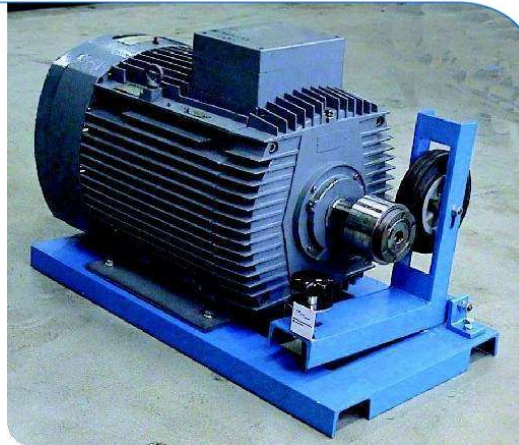
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Don't damage the motor during the conformance tests!

- Run the motor unbolted on an inertia base for baseline frequency data
- Running roller bearings without load can damage the bearings



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Before shipping, tie the motor shaft down

- Prevent transportation damage by strapping the shaft
- The static vibrations during transport can damage ("false brinelling") the bearings
- Motors with spring preloaded ball bearings usually do not suffer from this



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Prepare motor for transportation and storage

- Open motors should be wrapped to protect them from dust, dirt, and humidity
- Shaft-ends and flanges should be treated with an anti-corrosive
- Store in a clean, dry, and vibration-free environment
- Turn shafts every quarter



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Module closeout

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Name: _____ <Optional to Include Your Name>

WE 202 - Bearings in Rotating Machinery Applications
XX/XX/2007

Module Number & Title: 2. Bearings in Electric Motors

My **most important question on this module** is:
.....
.....
.....

I **did not understand** the following:
.....
.....
.....

I **do not agree with** the following:
.....
.....
.....

My **top three learnings from this module (referenced to your expectations)** are:

1.
.....
.....

2.
.....
.....

3.
.....
.....

Is this learning related to
your own expectations?

Yes	No
Yes	No
Yes	No

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